

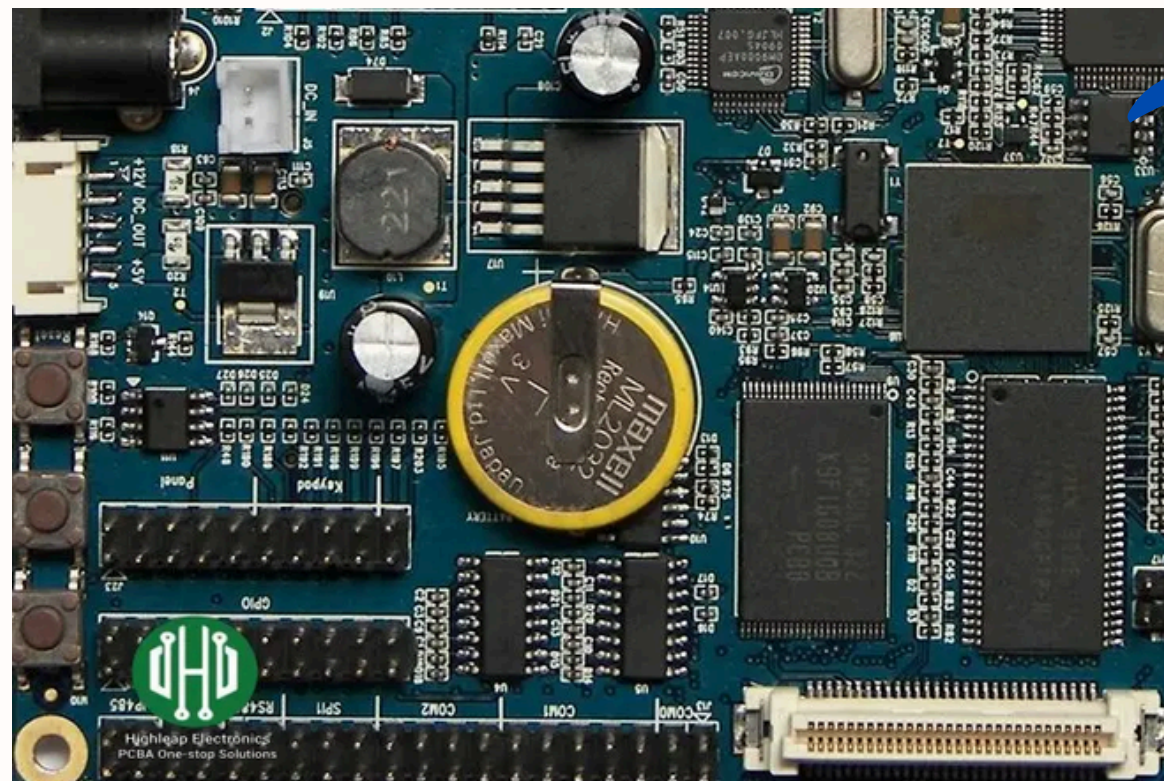


EGCI 232 Engineering Electronics

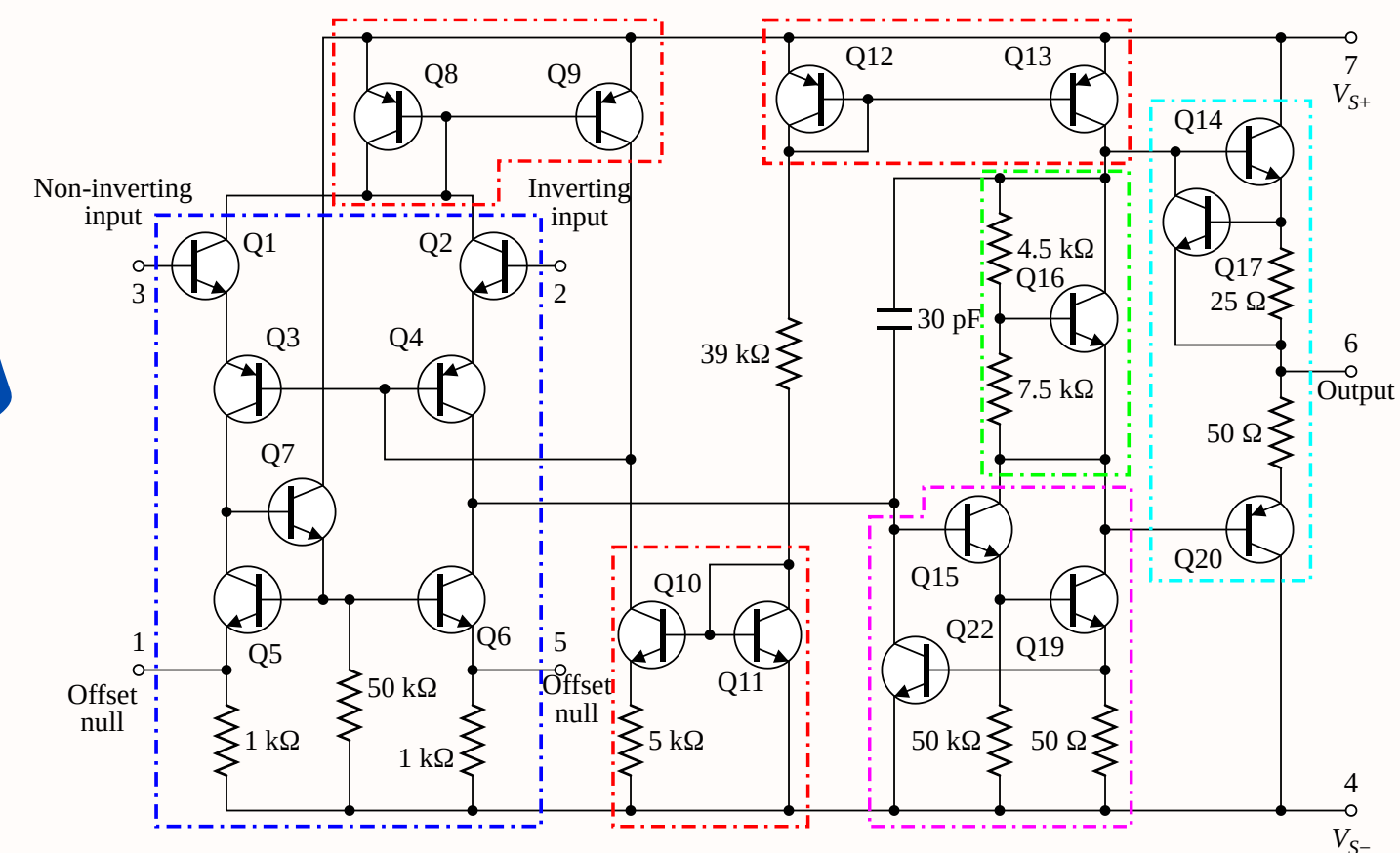
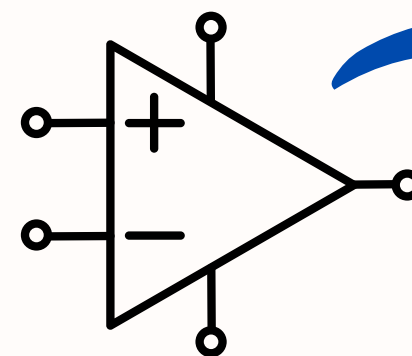
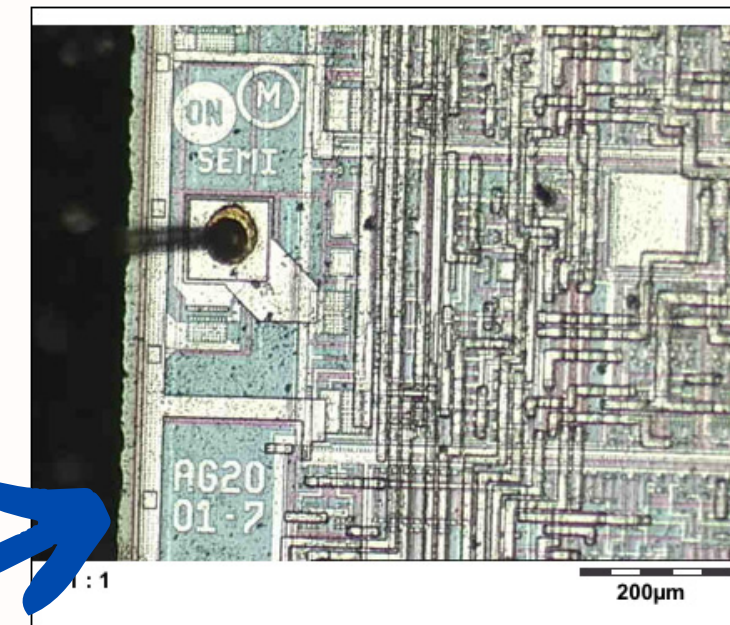
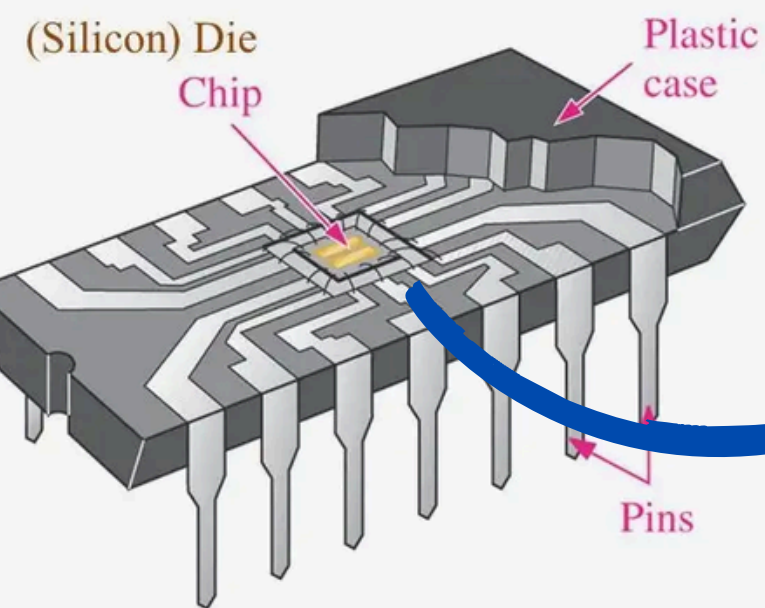
- Physics of Elements



Heart and brain of circuits

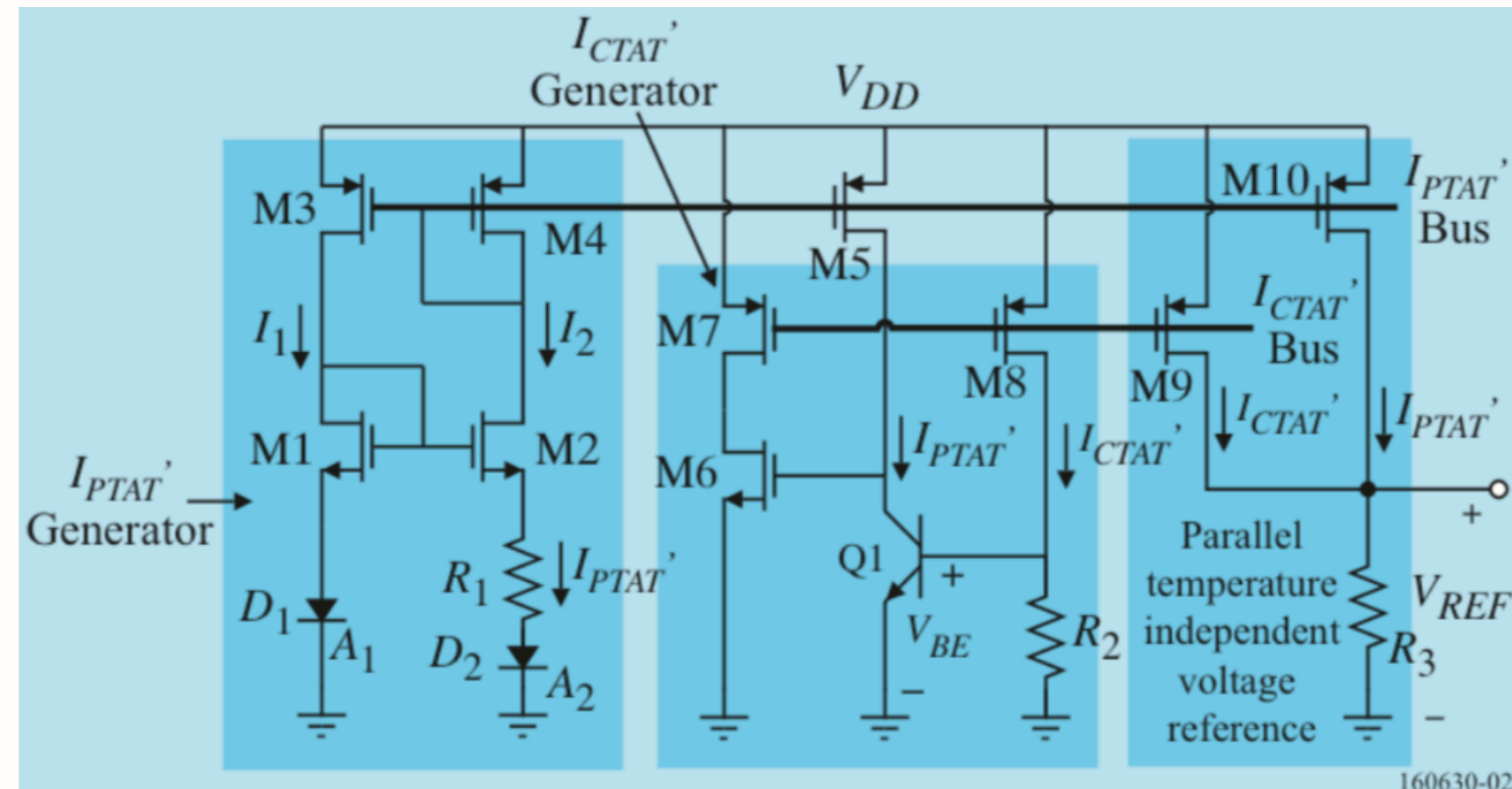


An Integrated Circuit

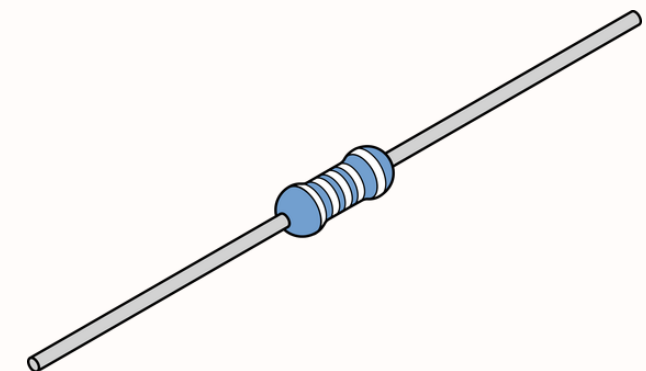
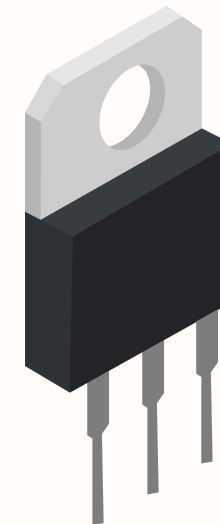
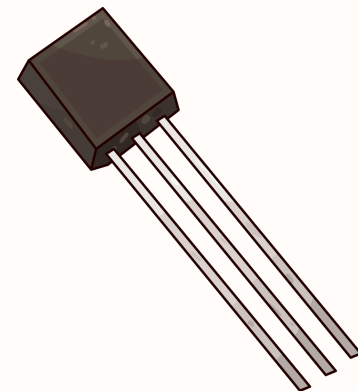
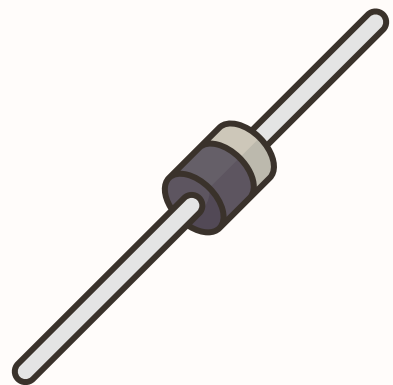




Let's get back to the basic



Ramdom example - CMOS





PERIODIC TABLE OF ELEMENTS

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American Chemical Society

www.acs.org/outreach

Read more: <https://inquivixtech.com/metalloids-are-useful-as-semiconductors-in-modern-electronics/>

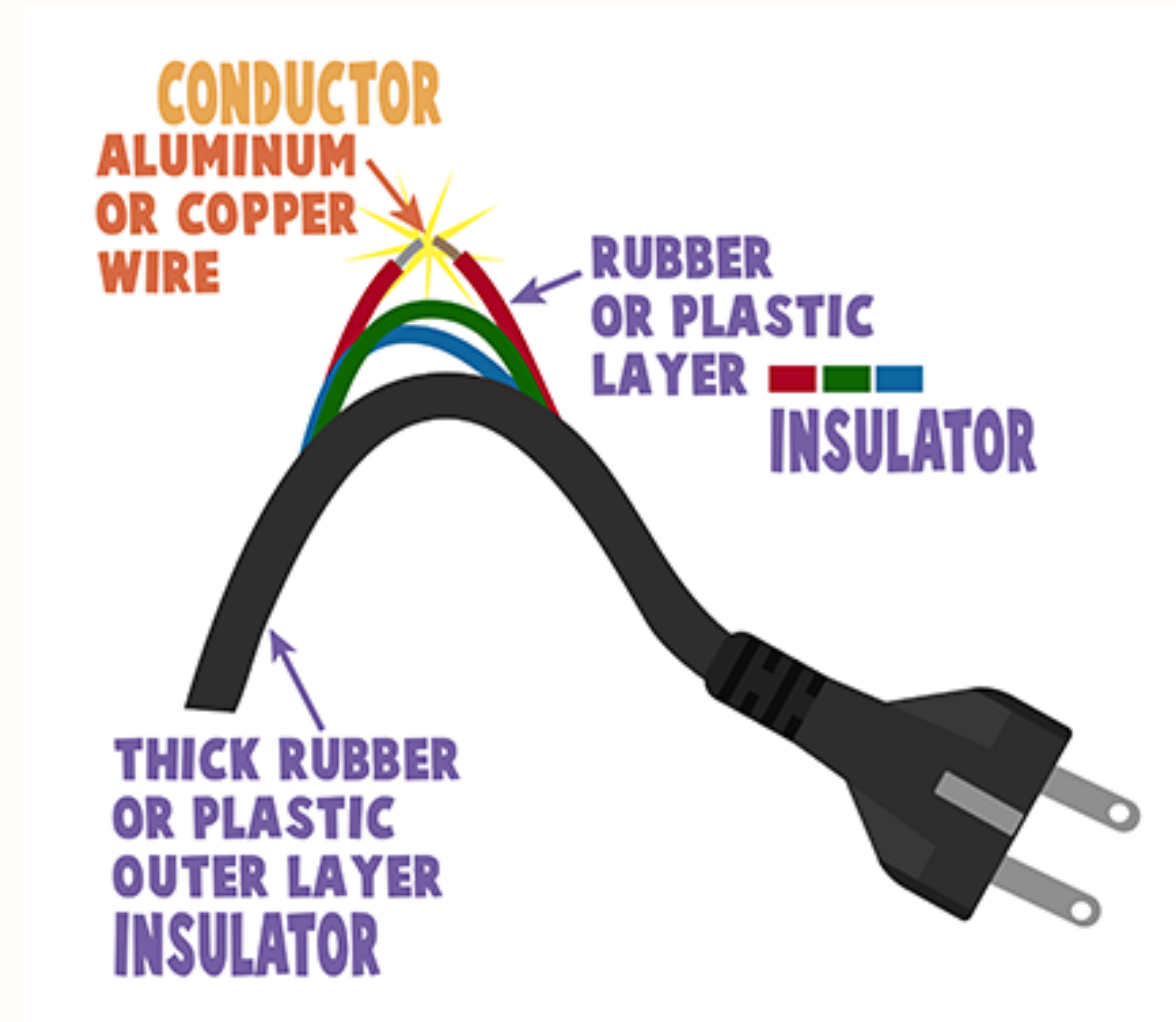


What are semiconductors ?

Semiconductors are materials whose electrical properties lie between Conductors and Insulators.

- ⚡ Materials that allow the electric current to pass through it.
- ⚡ Electrons move freely within the conductor.
- ⚡ Resistance is very low.

Conductors



- ⚡ Materials that do not allow the electric current to pass through it.
- ⚡ Electrons do not move freely within the insulator.
- ⚡ Resistance is very high.

Insulators

What are semiconductors ?

- Substances that conduct electricity under certain conditions.
- Pure materials (intrinsic) show no conduction behavior and also not insulation behavior.
- Impured materials (extrinsic) increase degree of conductivity.

Element(s)	Material
Single	Si, Ge
Compound	Group II-VI ZnSe CdTe
	Group III-V GaAs InGaN InP
	Group IV-IV SiC

Group II	Group III	Group IV	Group V	Group VI
Be Beryllium	B Boron	C Carbon	N Nitrogen	O Oxygen
Mg Magnesium	Al Aluminum	Si Silicon	P Phosphorus	S Sulfur
Zn Zinc	Ga Gallium	Ge Germanium	As Arsenic	Se Selenium
Cd Cadmium	In Indium	Sn Tin	Sb Antimony	Te Tellurium
Hg Mercury	Tl Thallium	Pb Lead	Bi Bismuth	Po Polonium



What are semiconductors ?

Electrical conductors



Silver



Gold



Copper



Steel



Sea Water

Electrical Insulators



Rubber



Glass



Oil

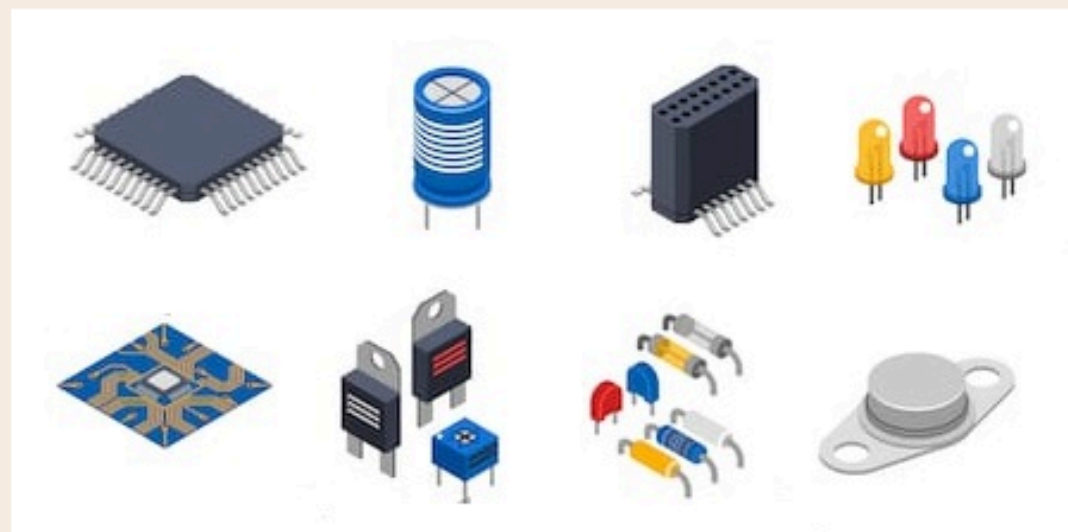


Diamond

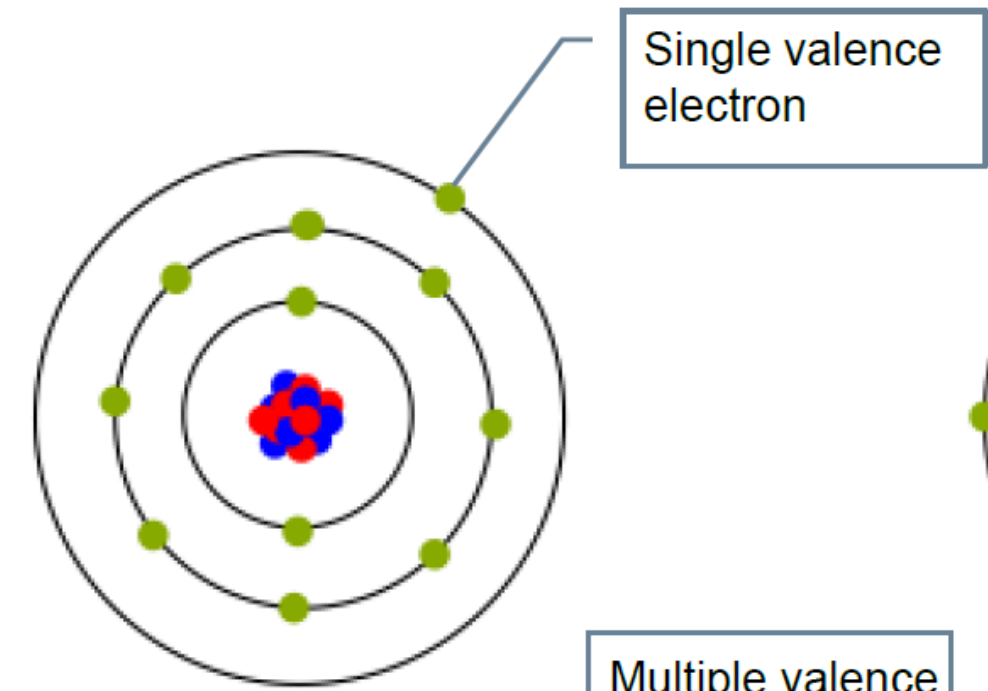


Dry wood

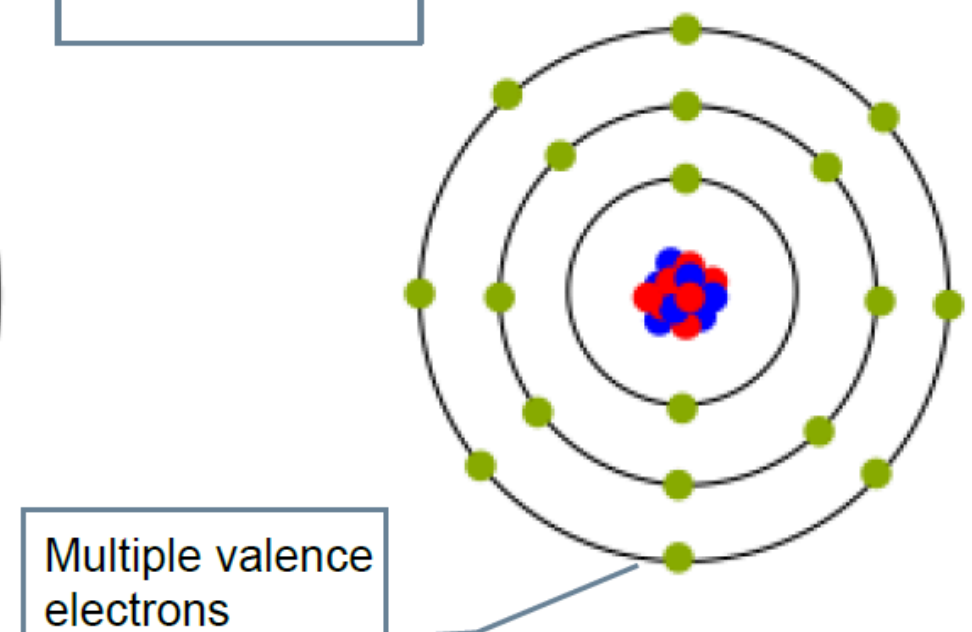
Semi Conductors



Conductor Atom

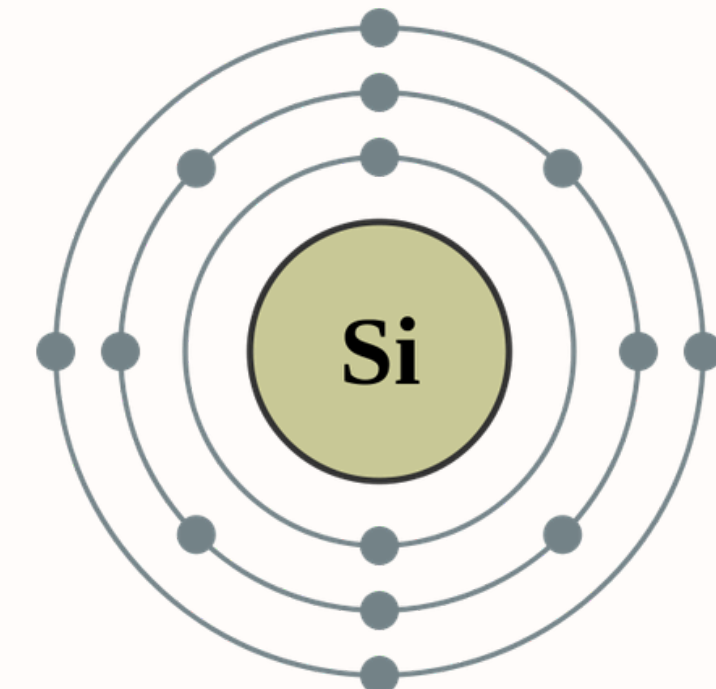


Insulator Atom

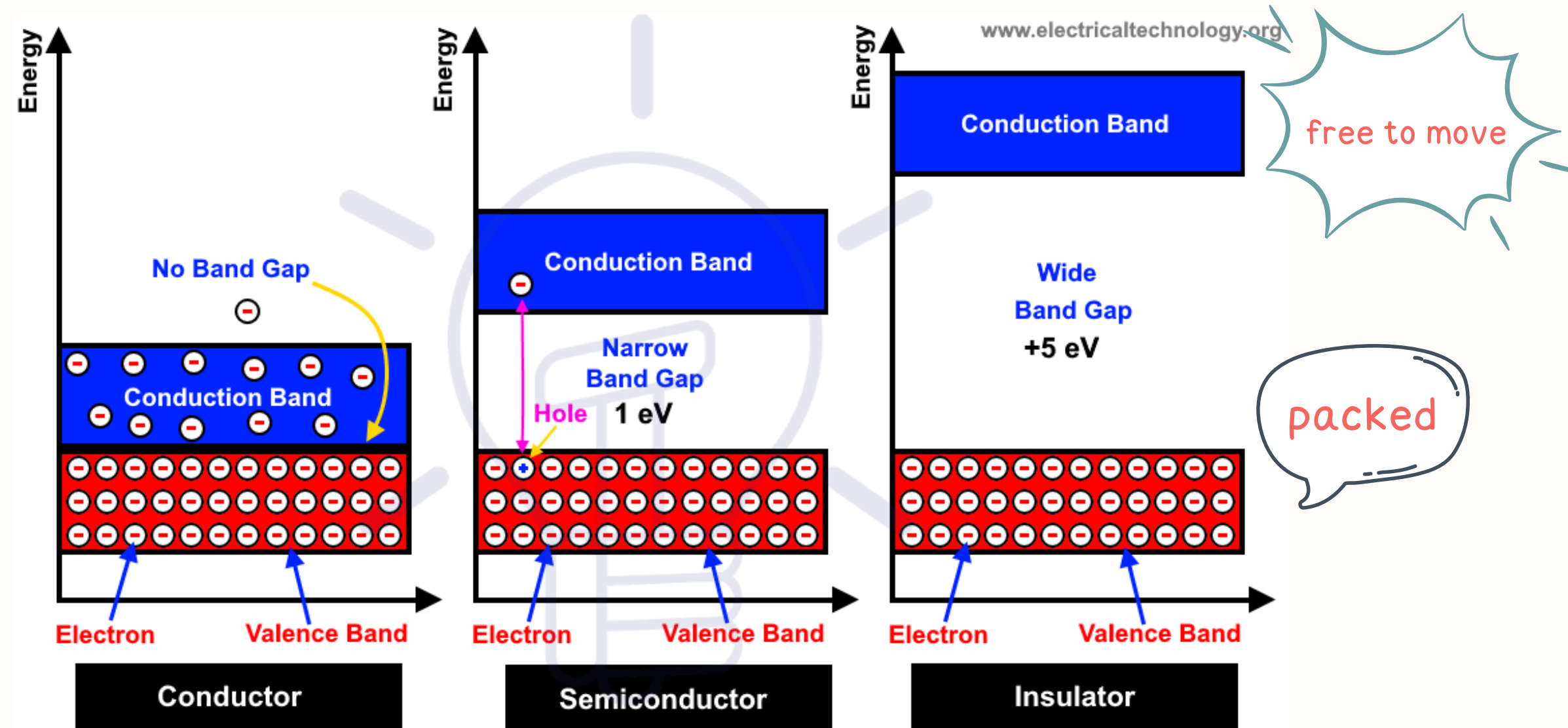


Semiconductor Atom

4 valence electrons



What are semiconductors ?

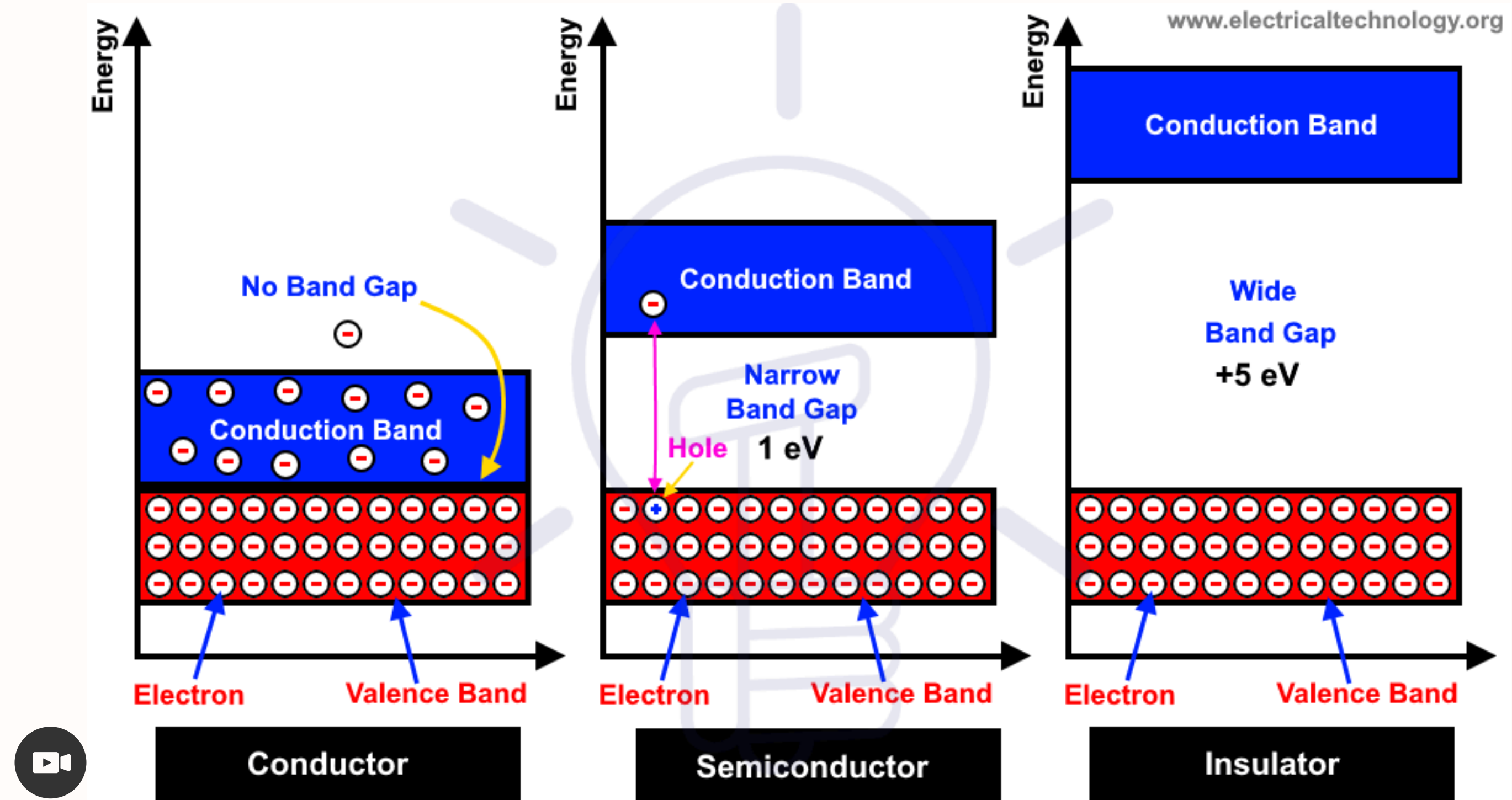


Material	Energy gap (eV) at	
	0 K	300 K
Si	1.17	1.11
Ge	0.74	0.66
InSb	0.23	0.17
InAs	0.43	0.36
InP	1.42	1.27
GaP	2.32	2.25
GaAs	1.52	1.43
GaSb	0.81	0.68
CdSe	1.84	1.74
CdTe	1.61	1.44
ZnO	3.44	3.2
ZnS	3.91	3.6

- **Valence Band:** The highest range of electron energies in which electrons are normally present at absolute zero temperature. Electrons in the valence band are tightly bound to atoms and are involved in forming chemical bonds.
- **Conduction Band:** The range of electron energies above the valence band where electrons are free to move and conduct electricity.
- **Bandgap:** The bandgap is the energy difference between the valence band and the conduction band.

What are semiconductors ?

- The band gap in semiconductors allows for the control of electrical conductivity through doping and temperature changes.



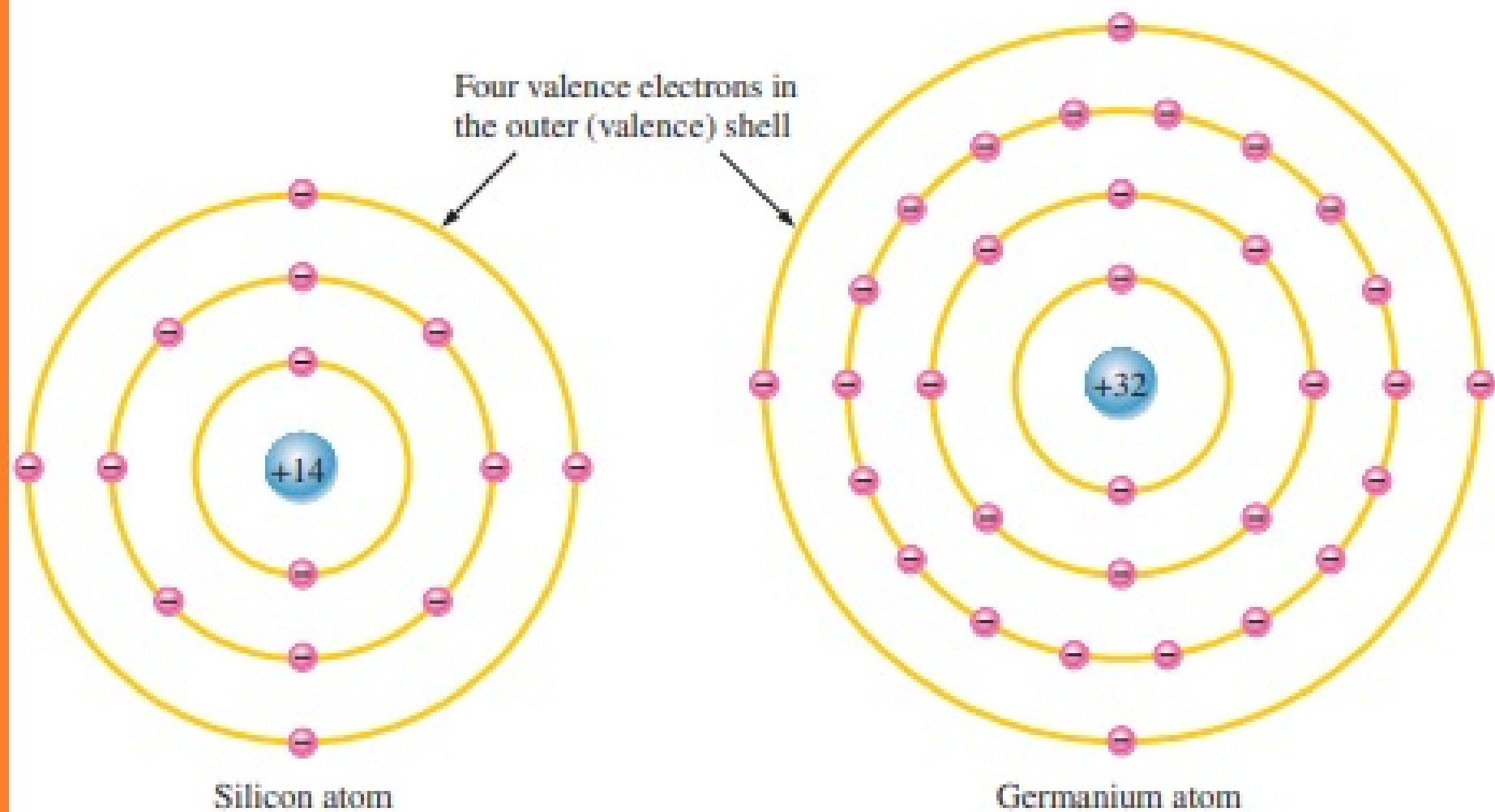
What are semiconductors ?

S.No	Conductors	Semiconductors	Insulators
1	Easily conducts the electrical current.	Conducts the electric current less than conductor and greater than insulator.	Does not conduct any current.
2	Has only one valence electron in its outermost orbit.	Has four valence electron in its outermost orbit.	Has eight valence electron in its outermost orbit.
3	Conductor formed using metallic bonding.	Semiconductors are formed due to covalent bonding.	Insulators are formed due to ionic bonding.
4	Valence and conduction bands are overlapped.	Valence and conduction bands are separated by forbidden energy gap of 1.1eV.	Valence and conduction bands are separated by forbidden energy gap of 6 to 10eV.
5	Resistance is very small	Resistance is high	Resistance is very high
6	It has positive temperature coefficient	It has negative temperature coefficient	It has negative temperature coefficient
7	Ex: copper,aluminium,etc	Ex: silicon, germanium, etc	Ex: Mica, Paper, etc



Silicon VS Germanium

Comparison between Silicon and Germanium atoms



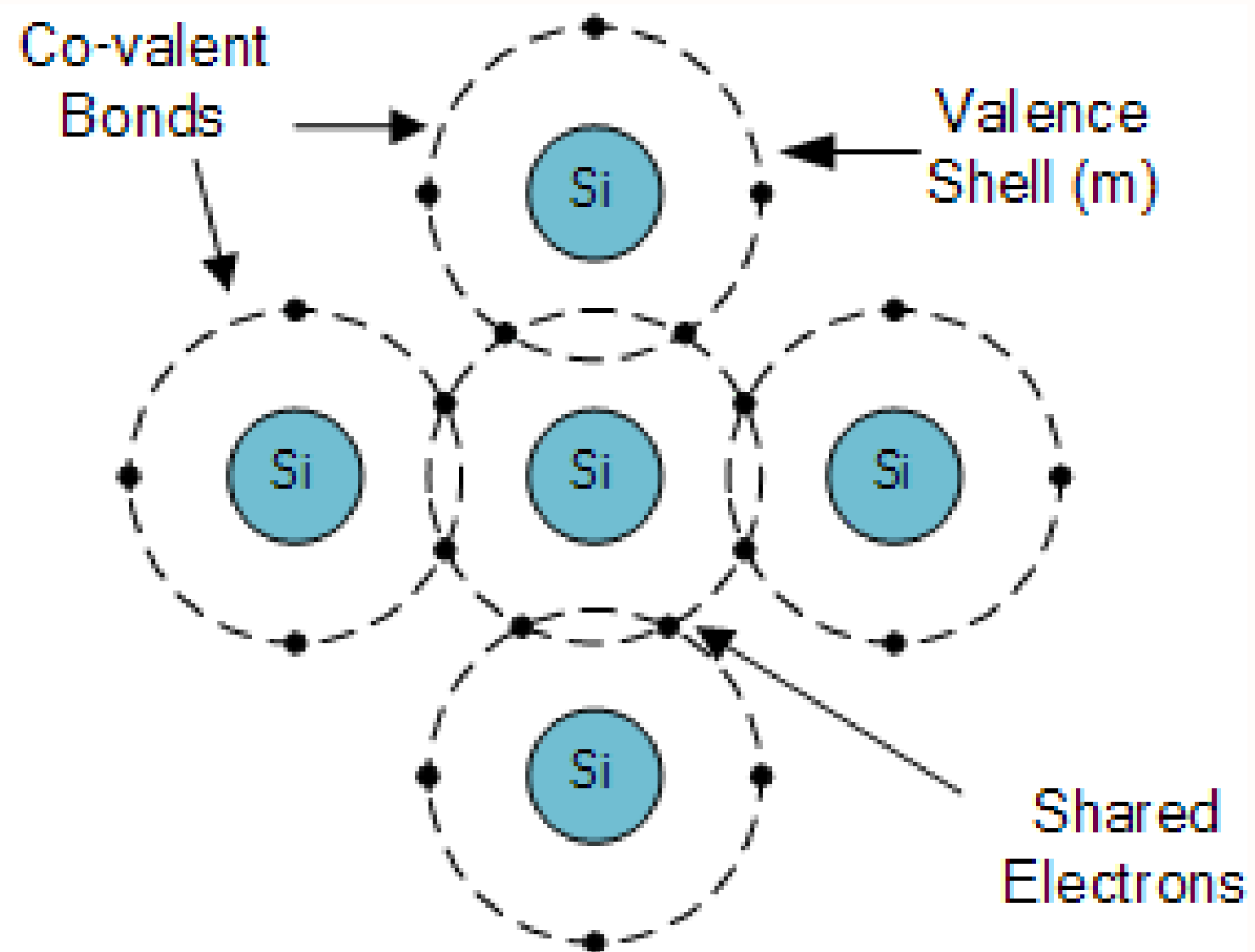
www.TheEngineeringKnowledge.com

- Silicon comes from silica sand - abundant supply.
- Another reason to choose silicon over germanium is thermal stability.
 - higher melting point.
- Germanium has less stable behavior
 - valance electrons in 4th outer shell experience less force of attraction so they could easily escape the outermost shell and get free.

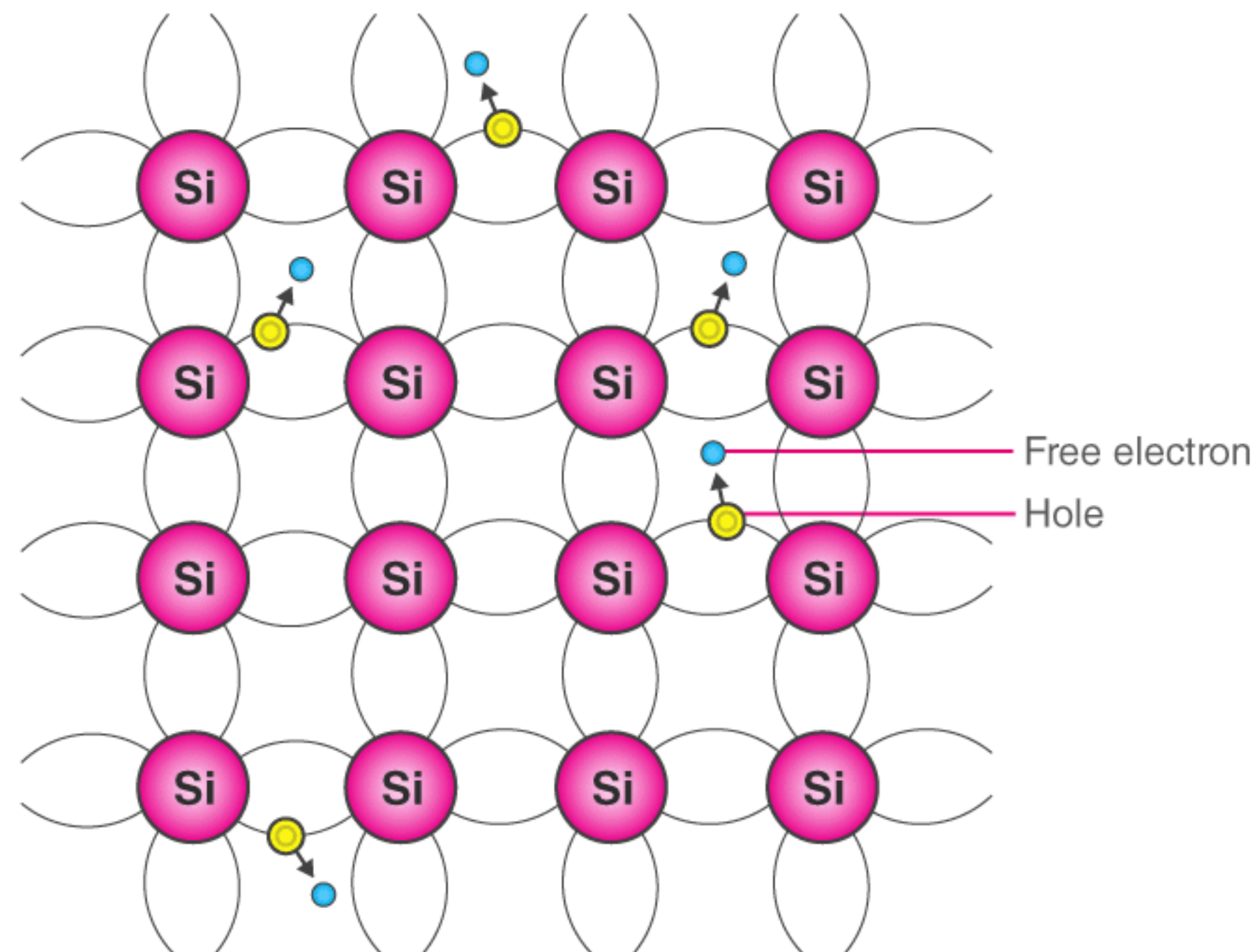


Intrinsic semiconductors

INTRINSIC SEMICONDUCTORS

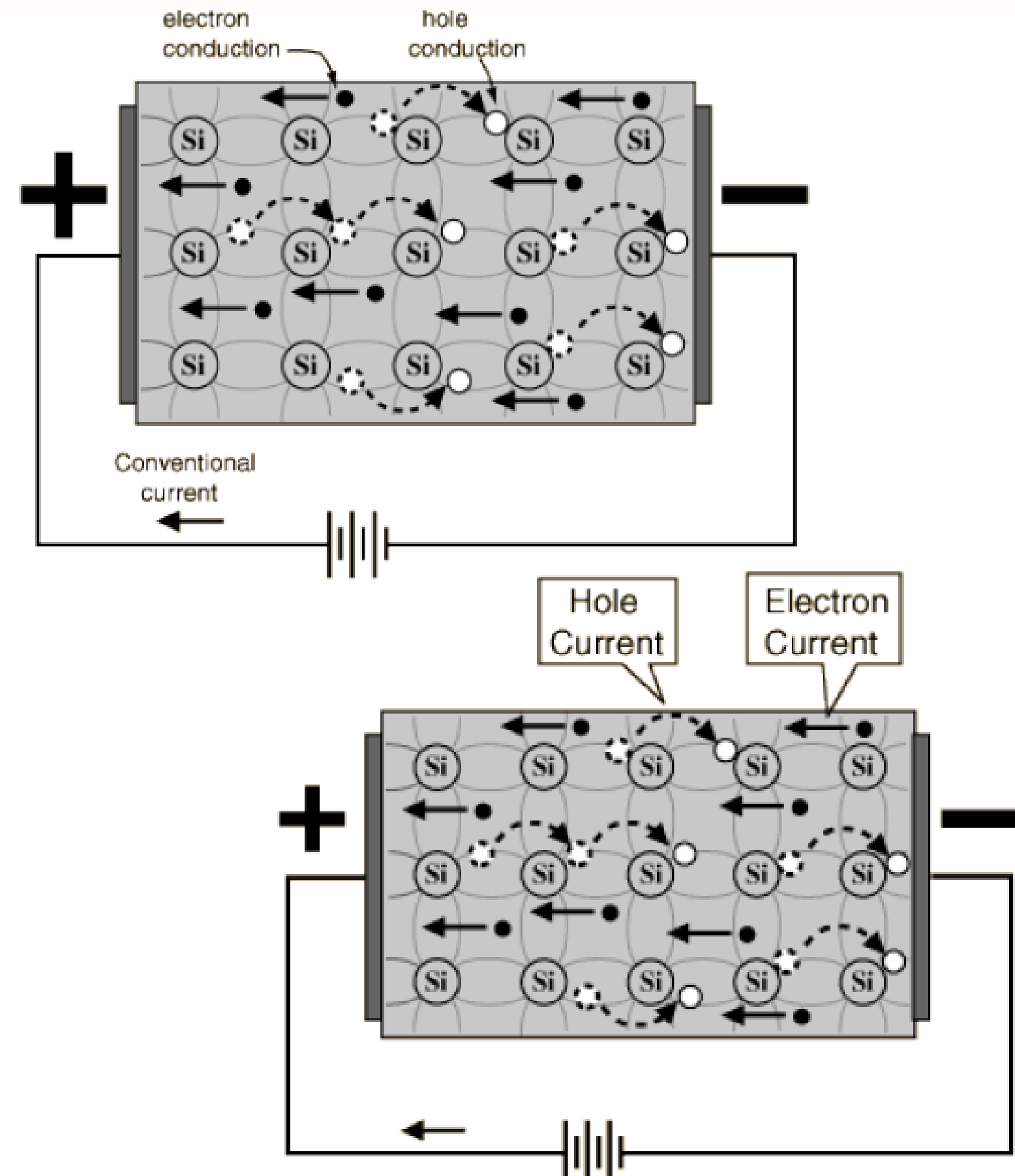


Silicon Crystal Lattice



Intrinsic semiconductors

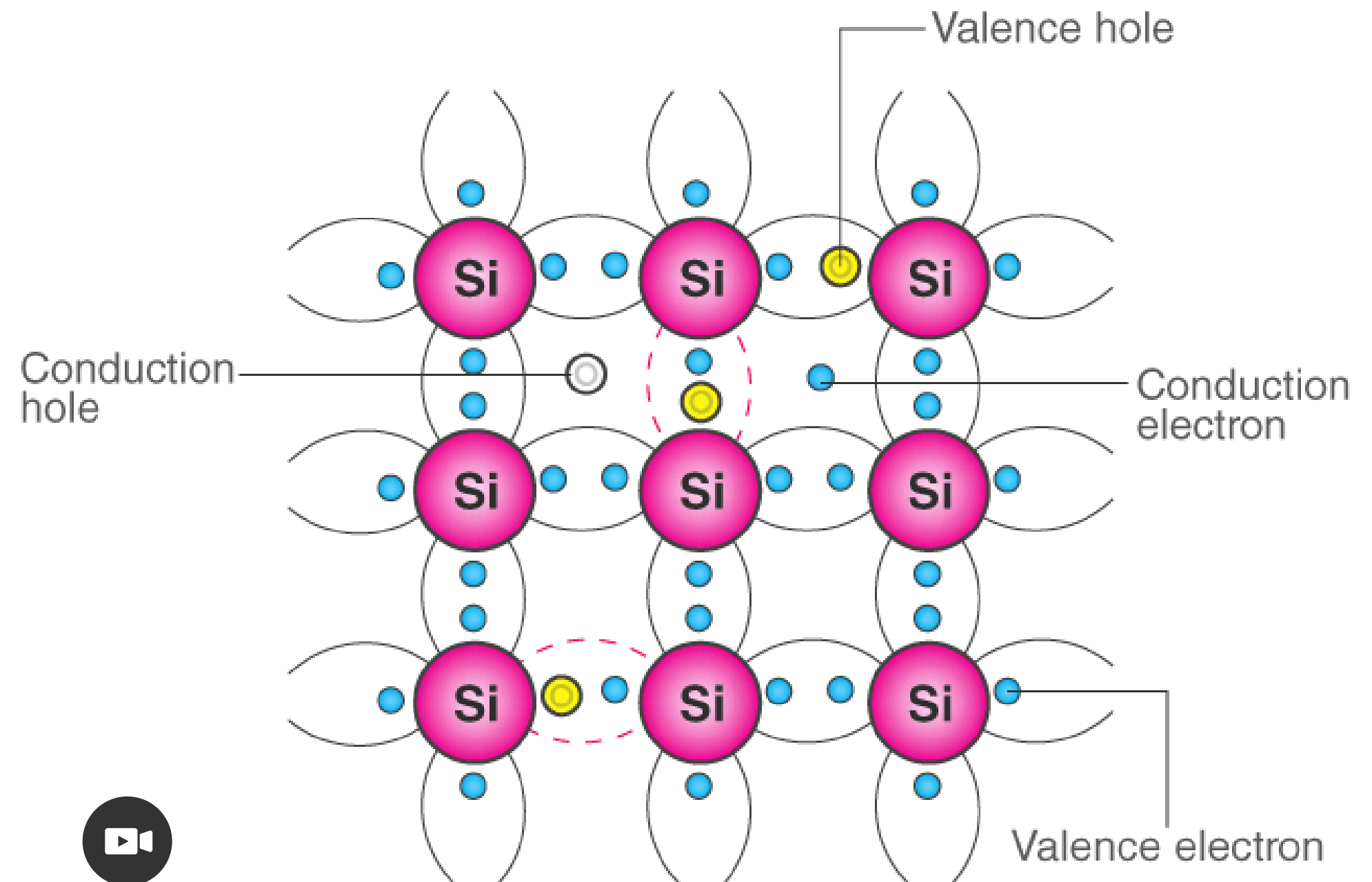
- If an electron is excited from the valence band into the conduction band (by thermionic processes, or incident radiation etc.), it leaves behind a 'hole' in the valence band.
- The 'hole' can move through the material in the valence band because electrons move to fill it, leaving a hole elsewhere.
- So both the 'hole' and the conduction band electron are free to move through the material. Their direction will be determined by a potential difference (holes to negative, electrons to positive).





Extrinsic semiconductors

- Pure silicon cannot conduct current well.
- By doping it with other substances:
 - increase the number of current carriers. Both electron and hole currents.
- Si + pentavalent atoms = N-type semiconductors
 - majority carriers are electrons.
- Si + trivalent atoms = P-type semiconductors
 - majority carriers are holes.



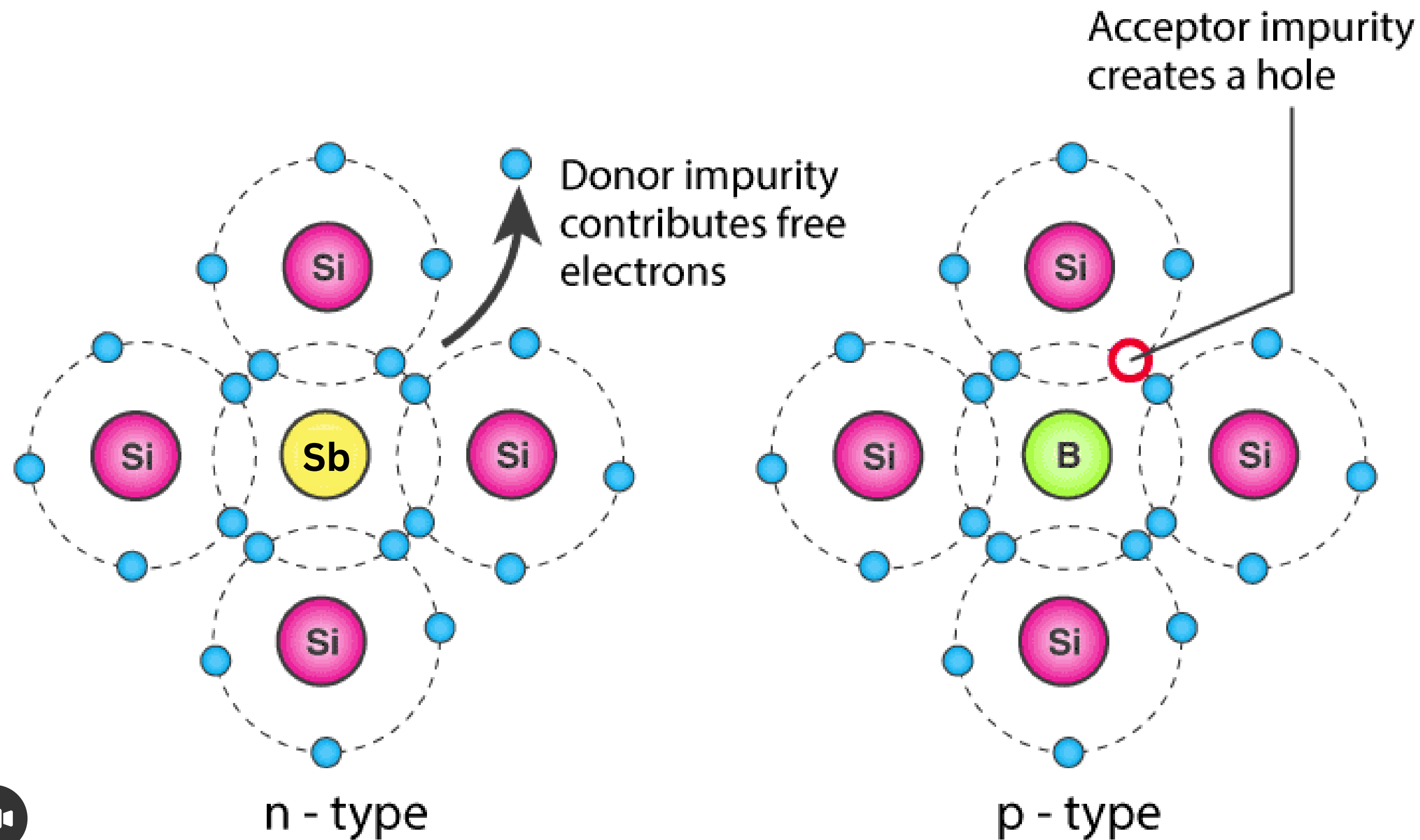


Doping

- A very common way of altering the electrical properties of a semiconductor is to add 'dopant' atoms into the uniform (silicon) crystal structure.
- The effect of doping is to introduce new energy levels between the conduction and valence bands.
- This is a bit like a 'ladder', helping electrons move between the two levels.
- The dopant atoms only have to be present in very small quantities to change the electrical behavior of the semiconductor (1 in 200,000 atoms). There are two types of doped semiconductors:



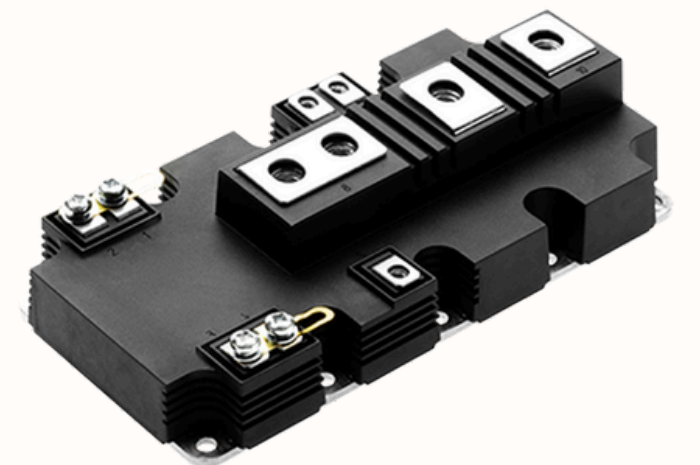
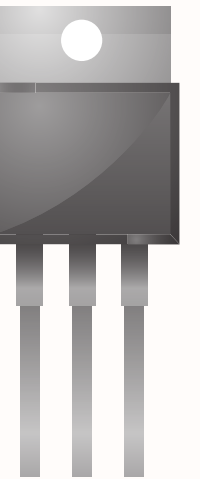
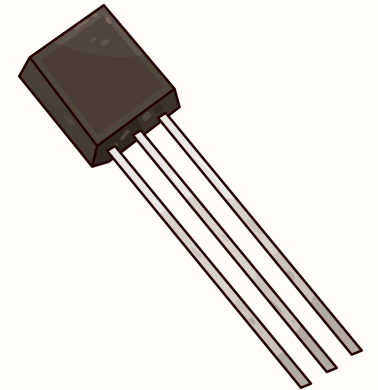
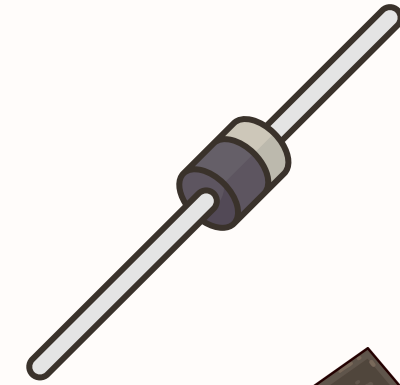
Extrinsic semiconductors





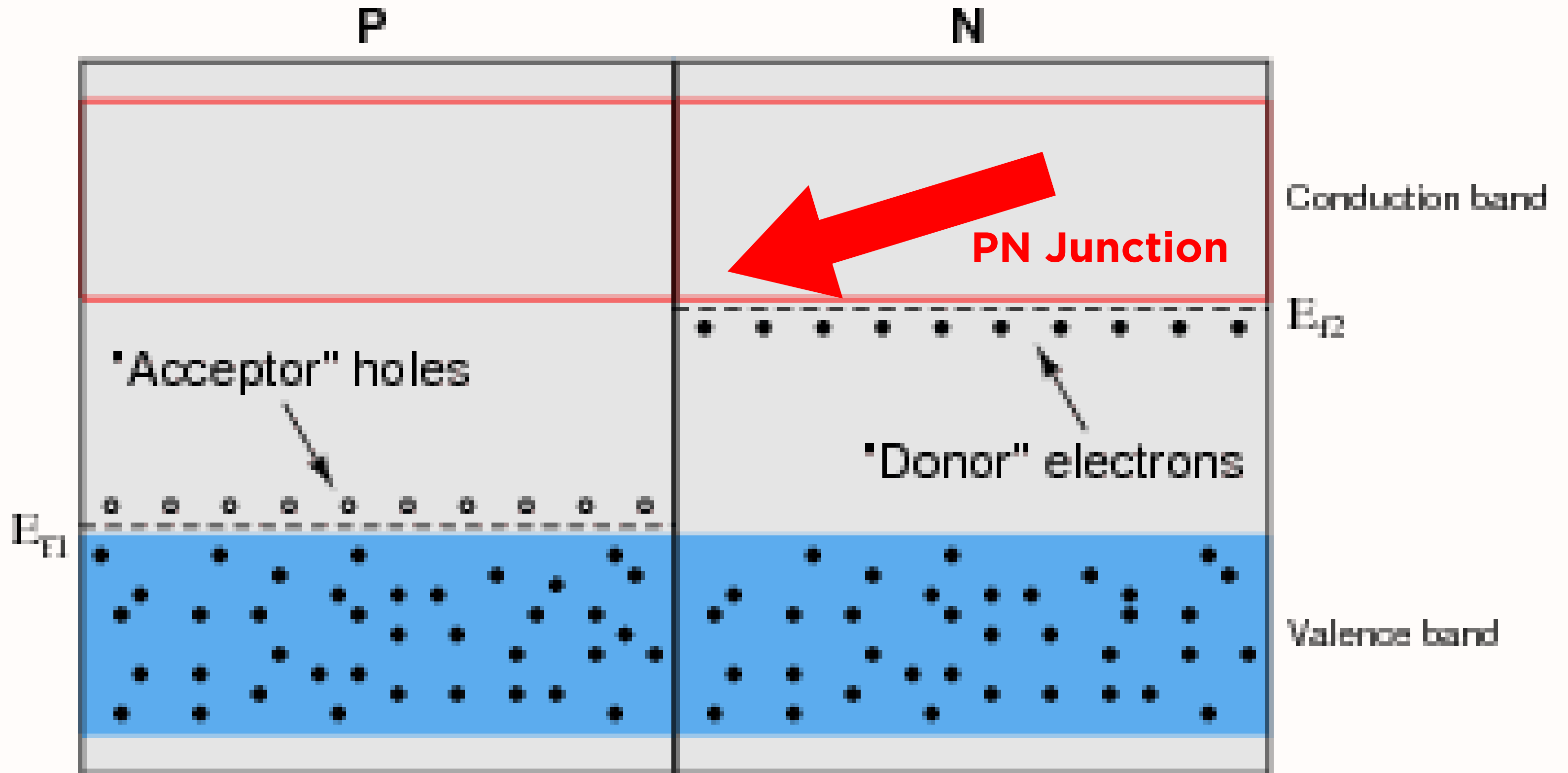
We are good to go!

- Diodes
- Bipolar Junction Transistors (BJTs)
- Metal Oxide Semiconductor Field Effect Transistors (MOSFETs)
- Thyristors
- Triacs
- Silicon Controlled Rectifiers (SCRs)
- Insulate Gate Bipolar Transistors (IGBTs)



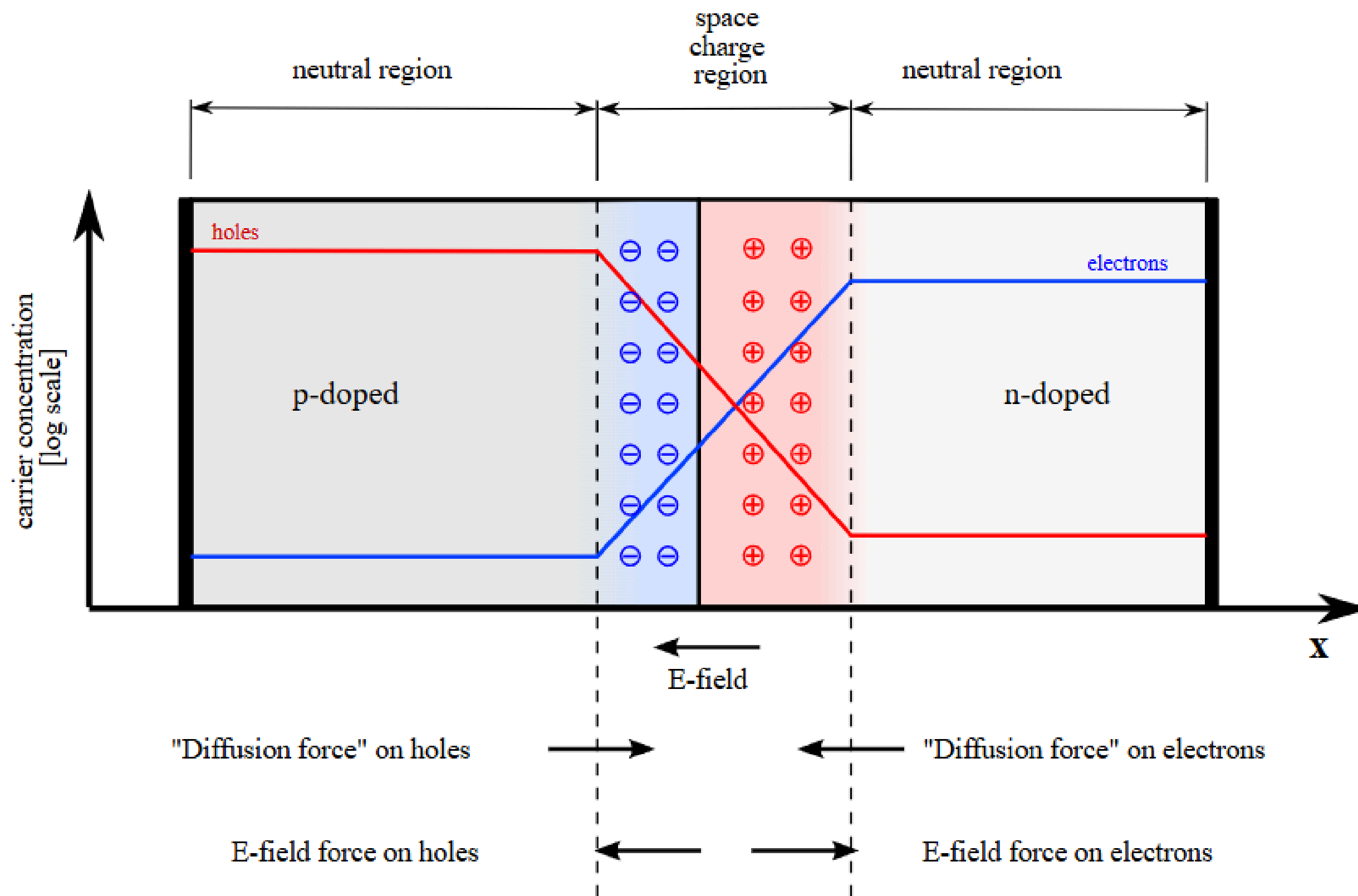
Basic Structure of electronic device

PN Junction - at 0 K (absolute zero)



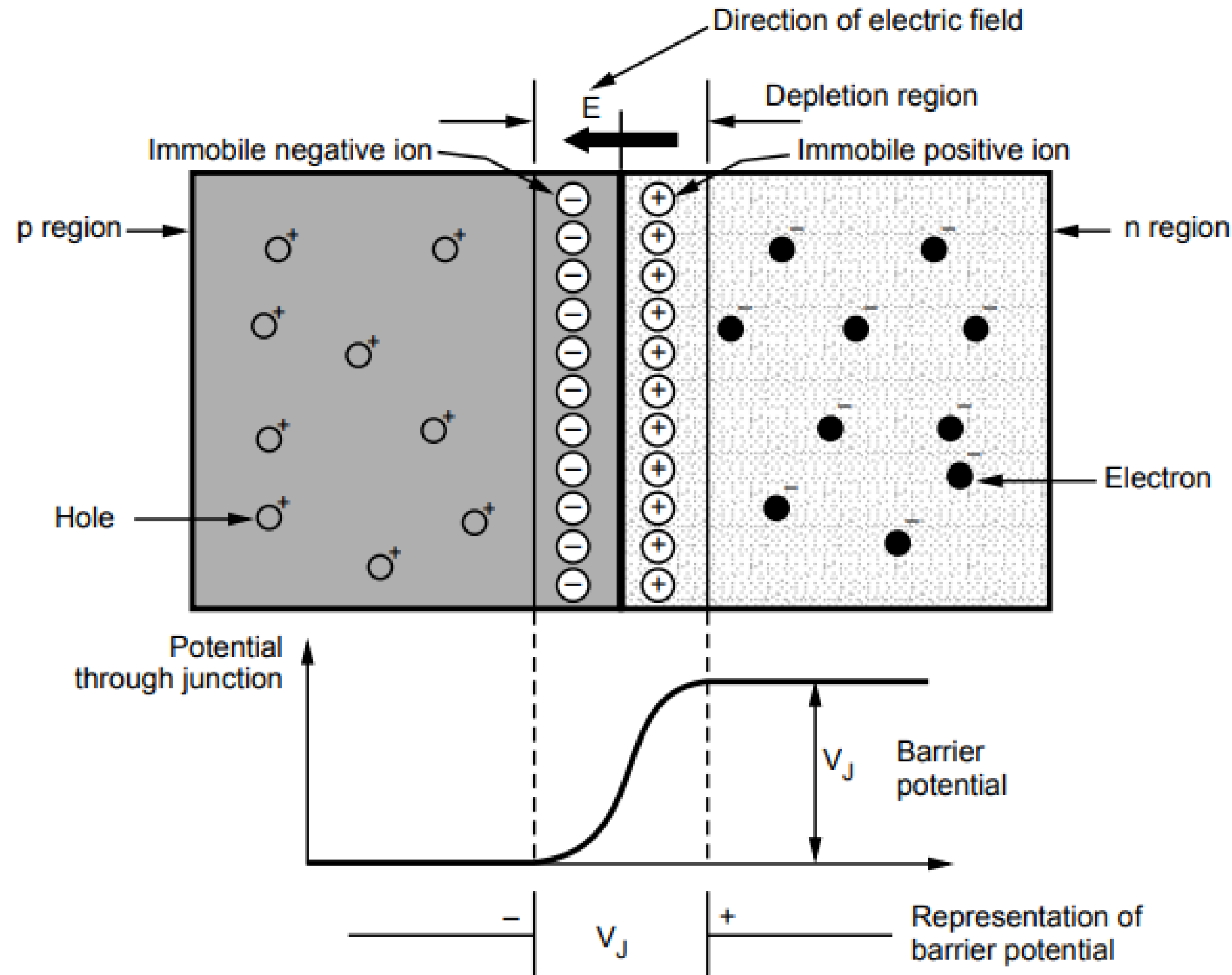
Basic Structure of electronic device

PN Junction - at room temperature



Basic Structure of electronic device

PN Junction - unbiased

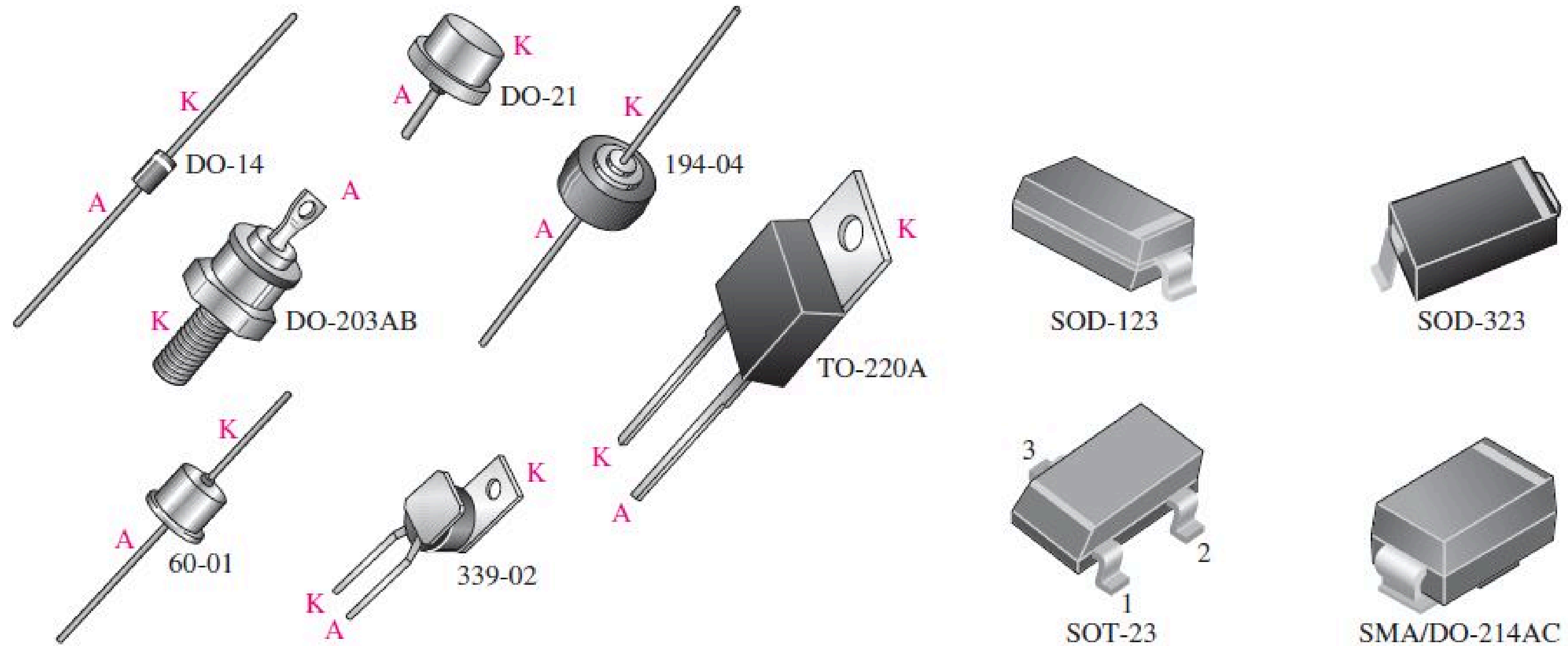
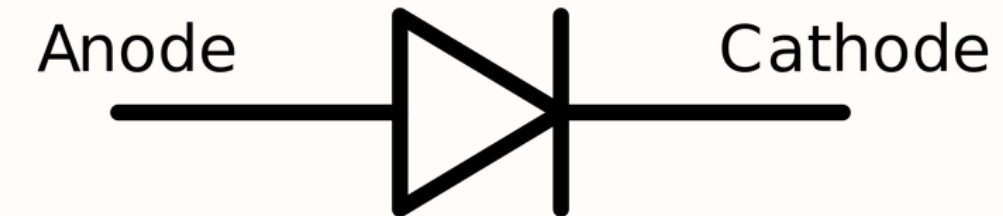


PN junction

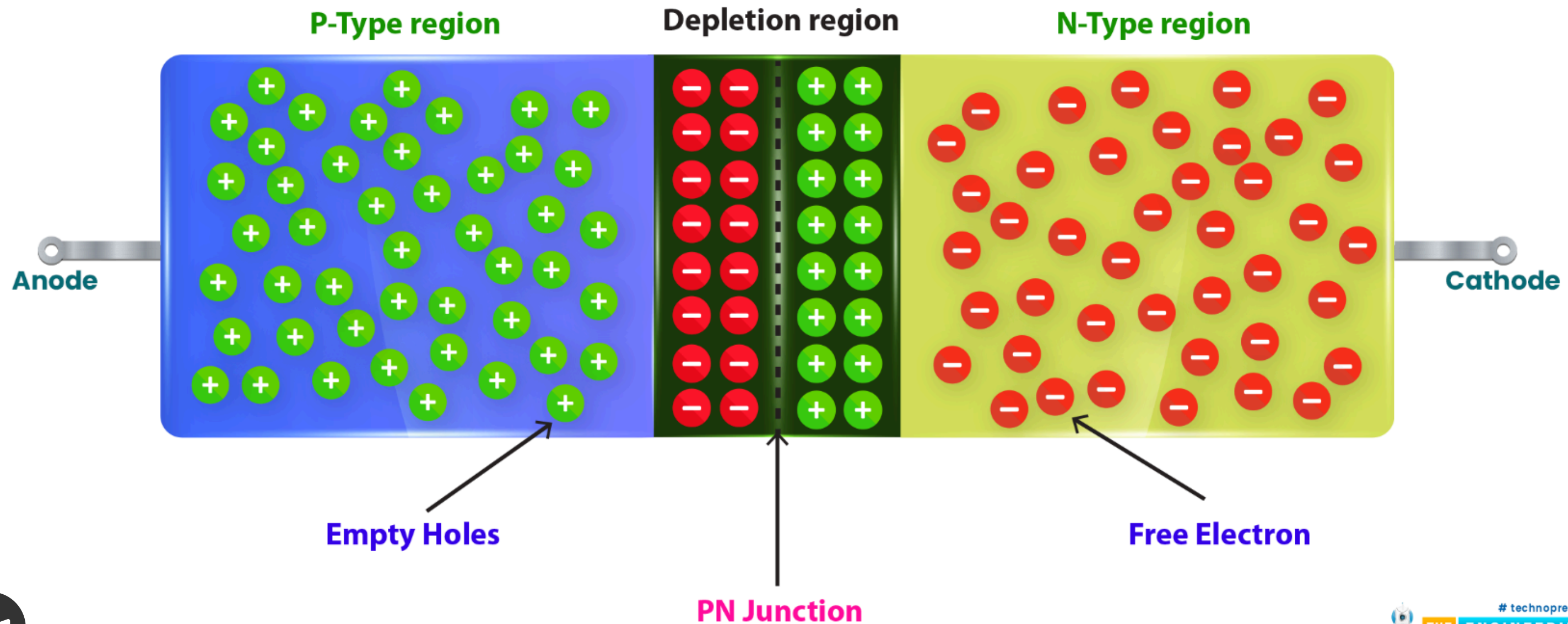
- Electric field is generated from the diffusion of carriers
- Electric field creates potential across the barrier
- Depletion layer (or space-charge region) is the area without carriers due to the recombination of free electrons and hole
- Electric field causes barrier potential, which prohibits major carriers to flow to the other side
- Approx. barrier potential of Germanium (Ge) is 0.3 V and 0.7 V for Silicon (Si)



Diode - the simplest electronic device

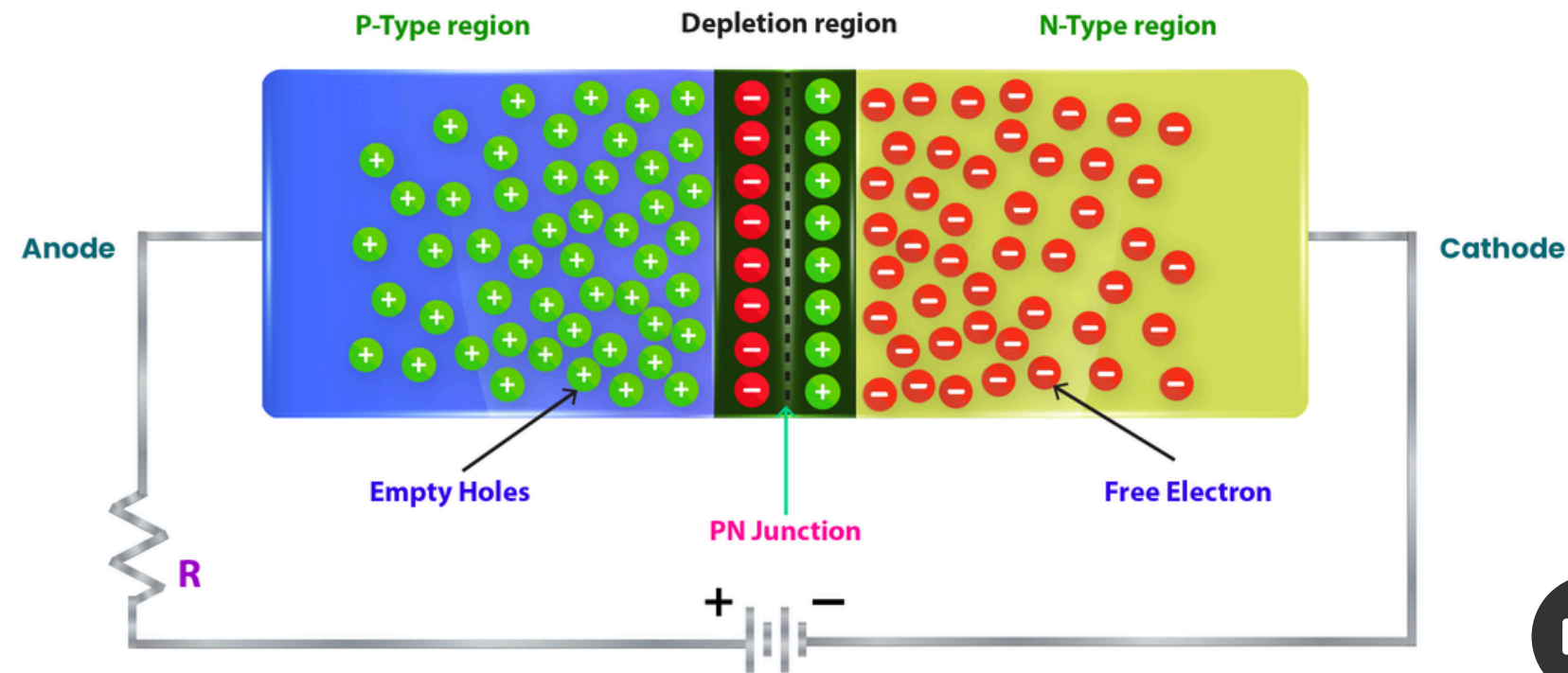


Formation and Properties of Junction Diode

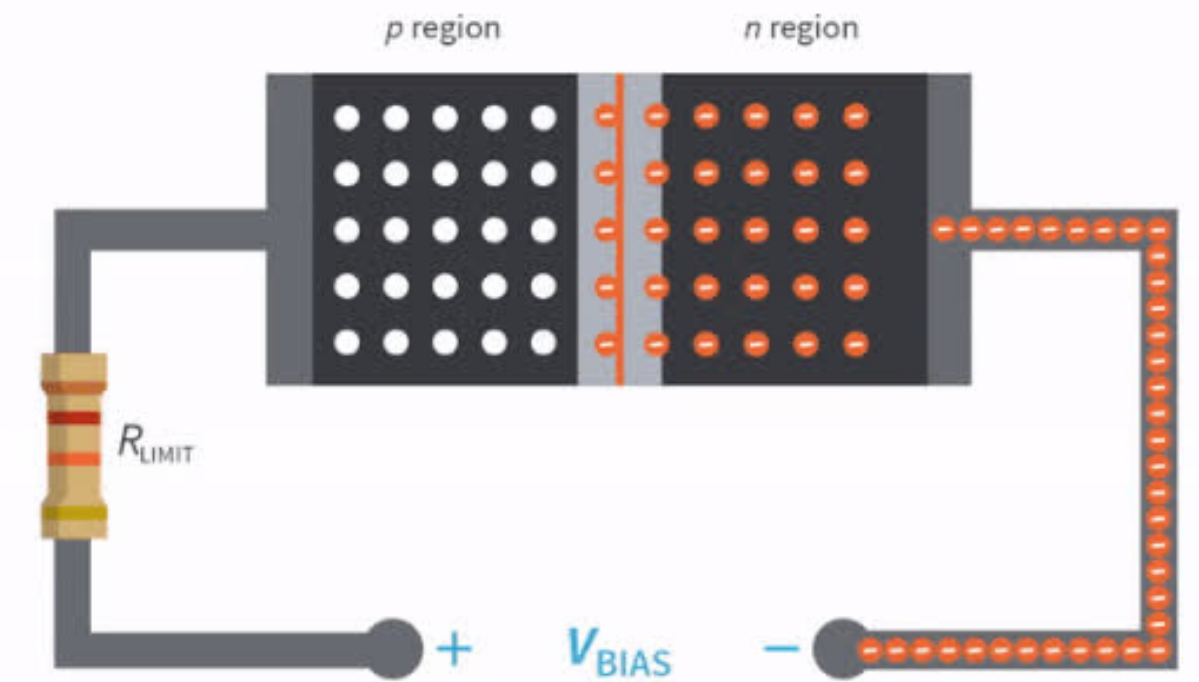


Working principles of diodes

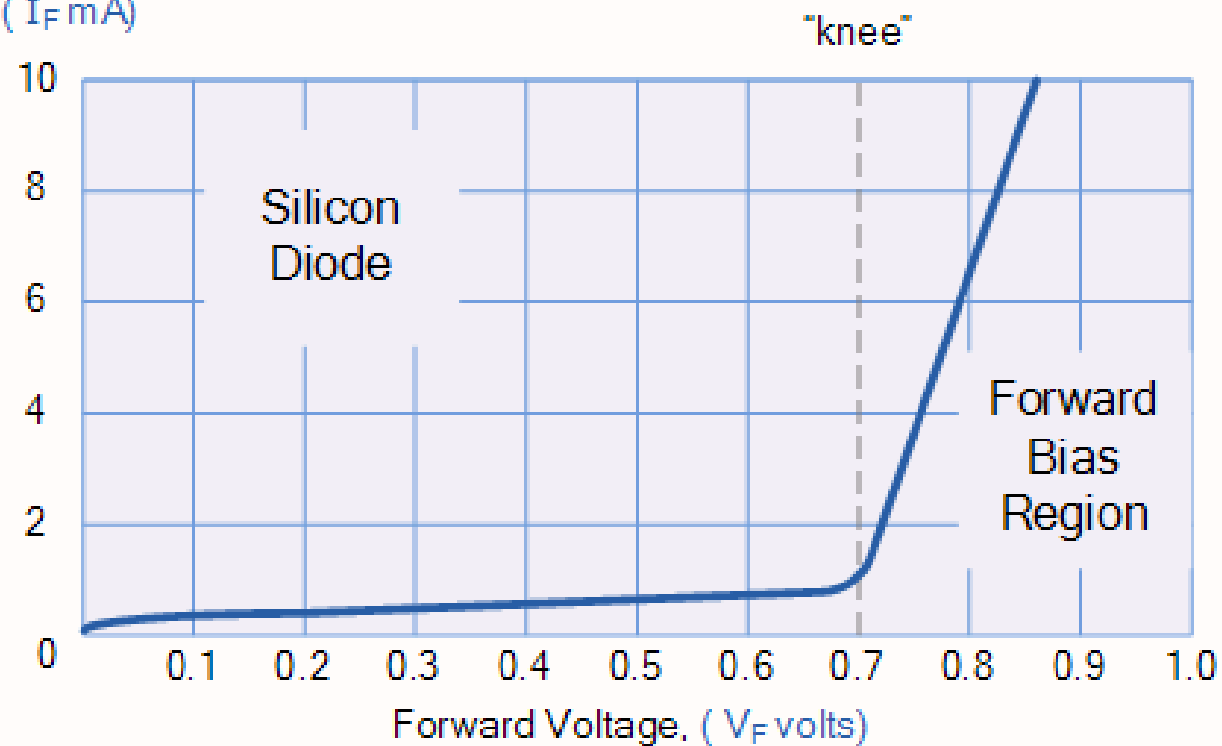
Forward-Biased PN Junction



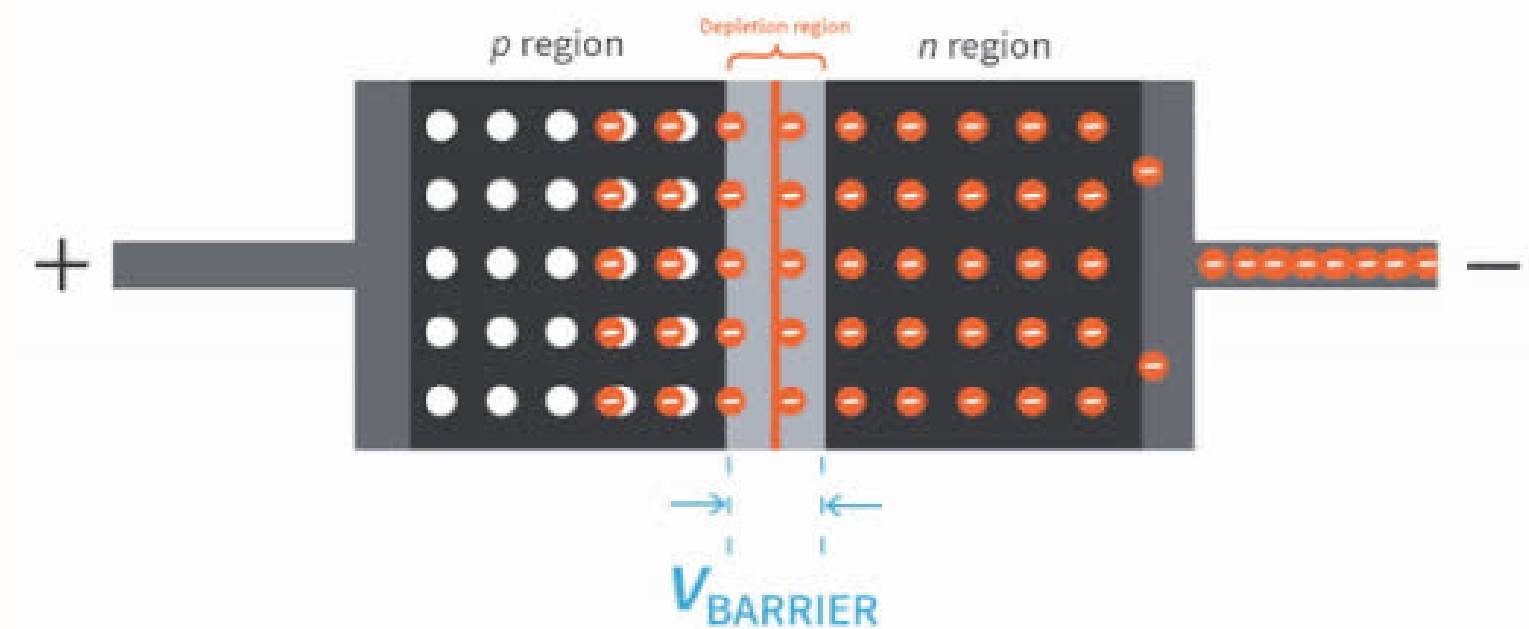
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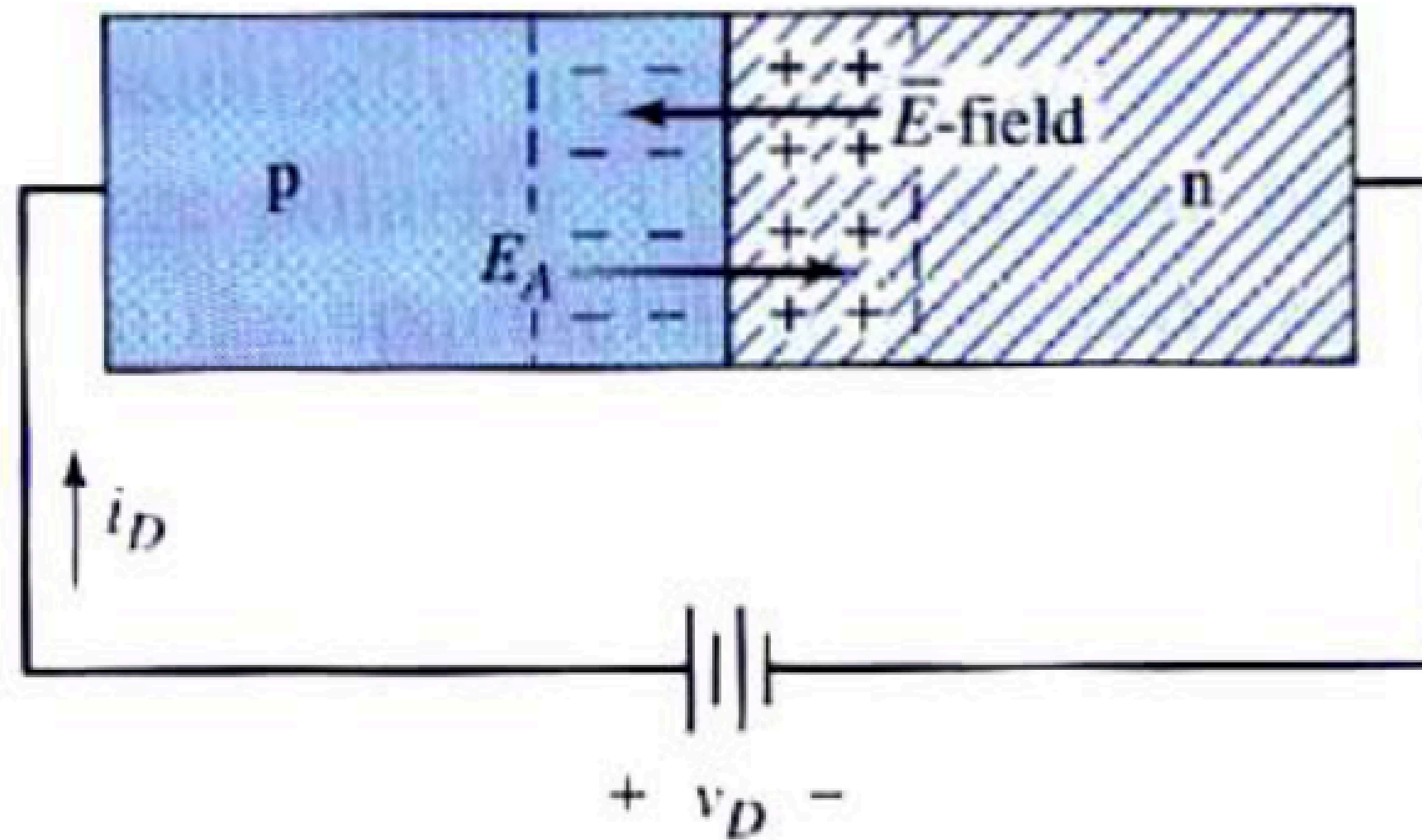
Forward Current
(I_F mA)



$$i_D = I_s \left[\exp\left(\frac{v_D}{nV_T}\right) - 1 \right]$$



Working principles of diodes



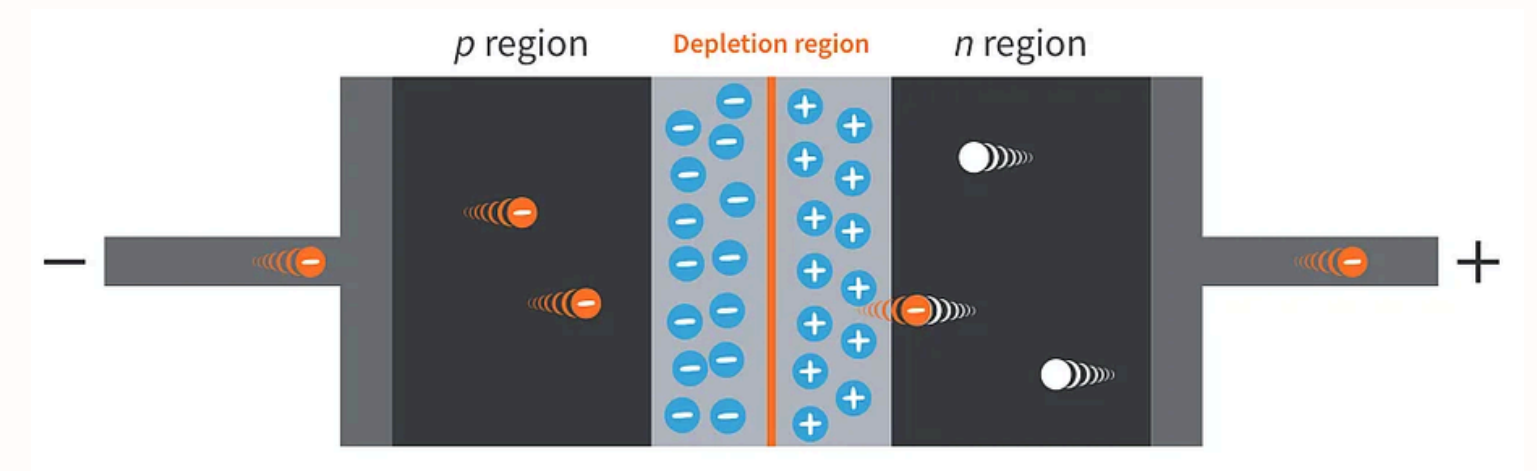
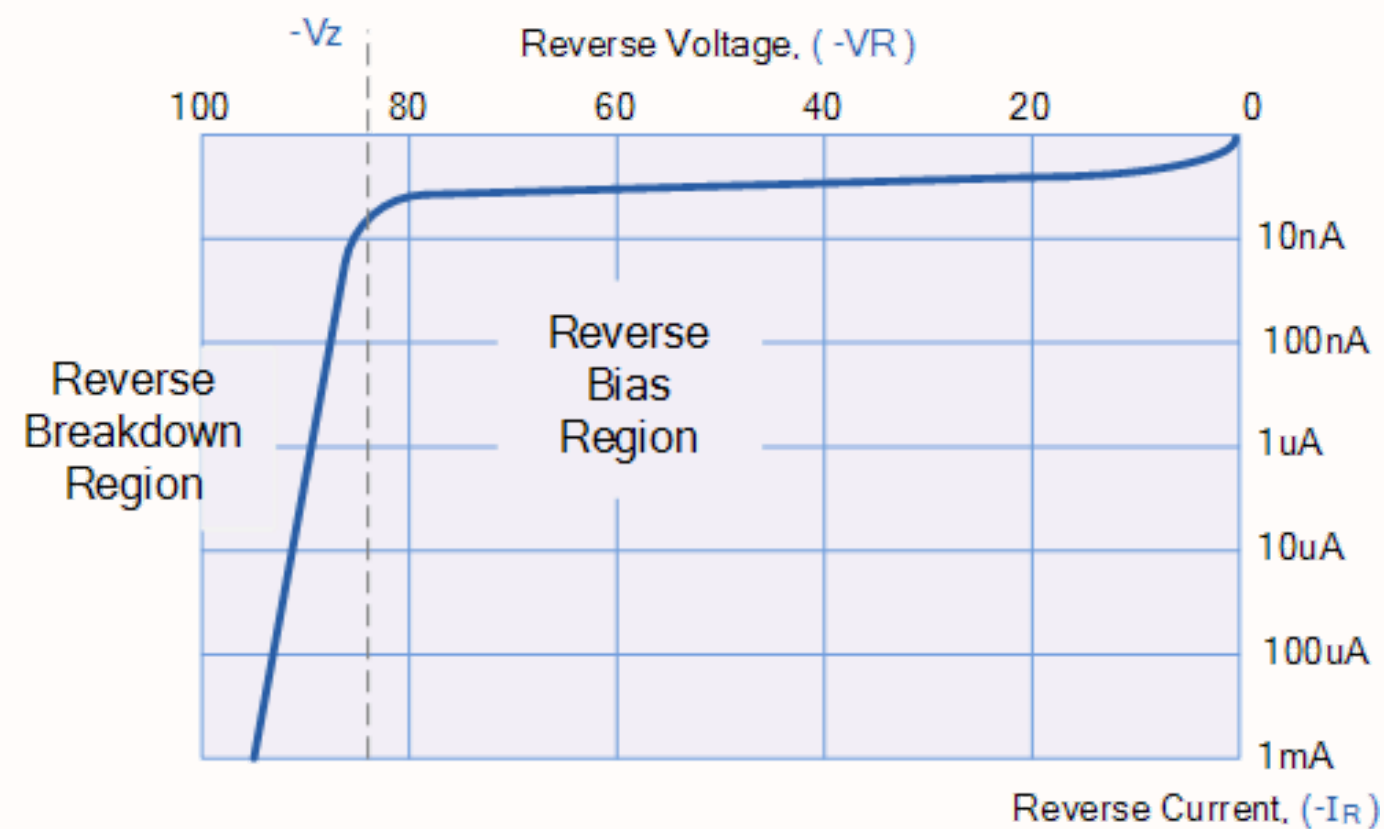
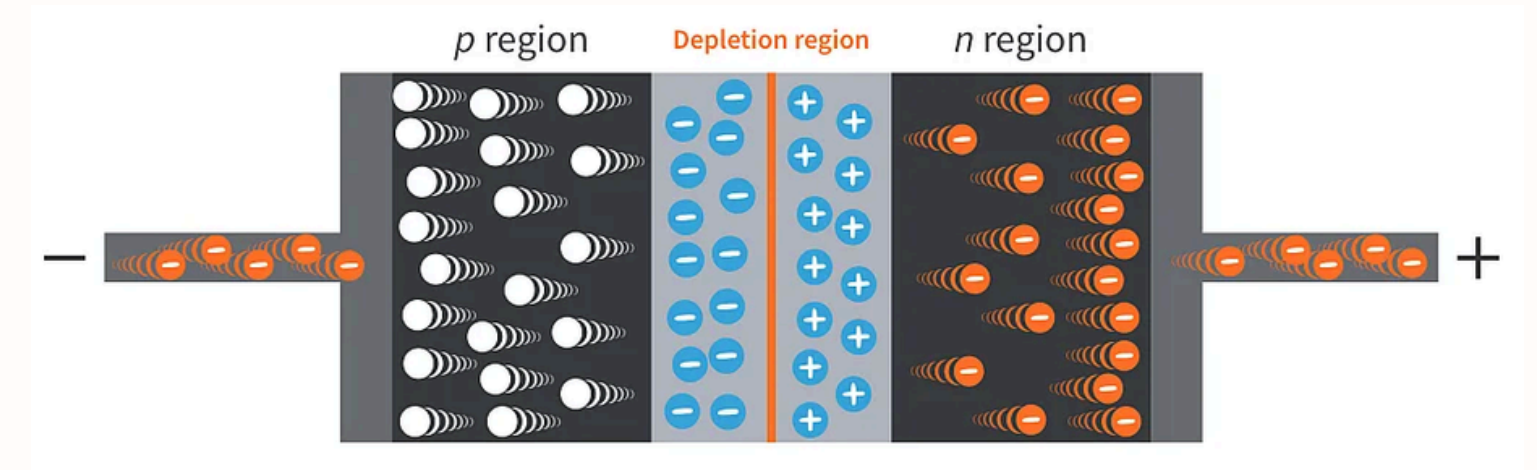
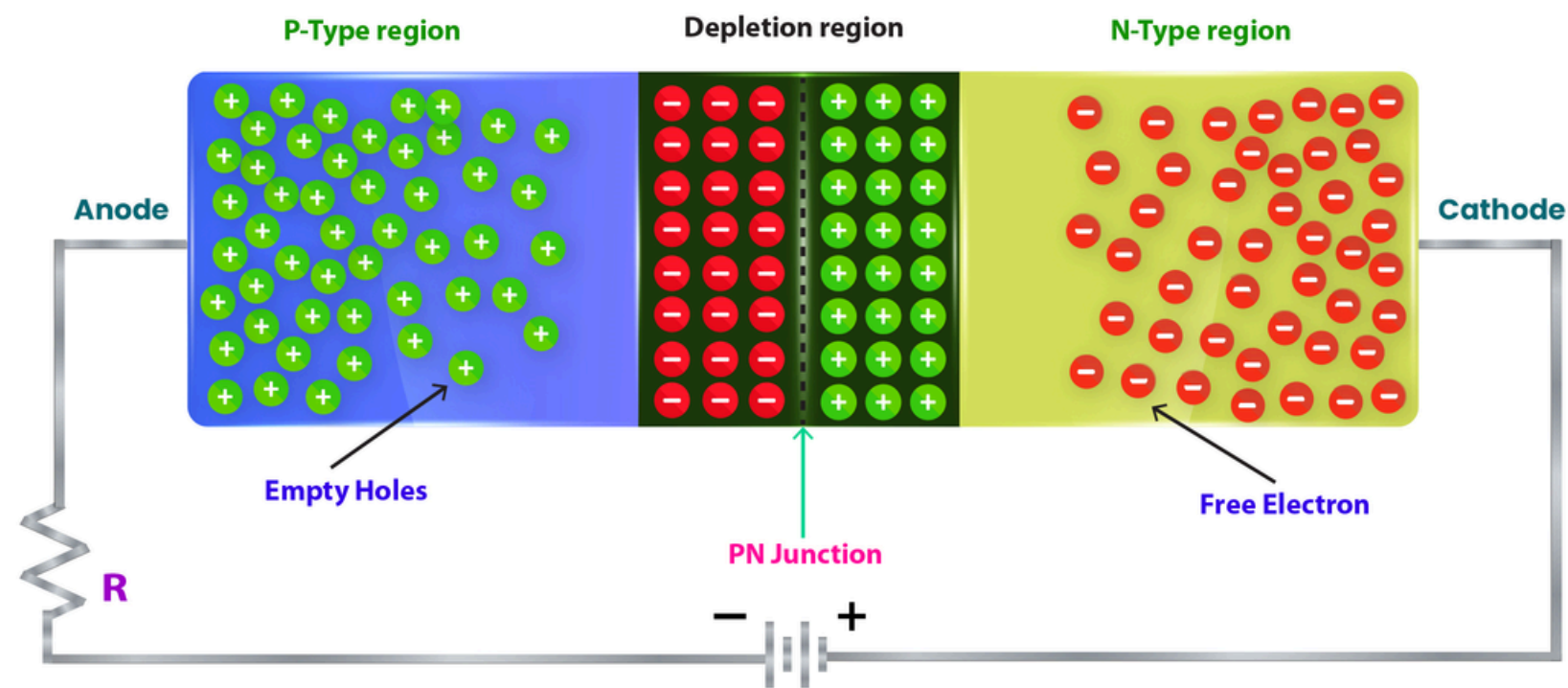
□ Forward-biased PN junction

- Net E-field is less than at the equilibrium, which results in narrower space-charge region and stirs the balance between diffusion and E-field force
- Majority carriers diffuse into the opposite side as long as V_D is applied and create current through the junction just like lowering a dam wall to let lots of water flows

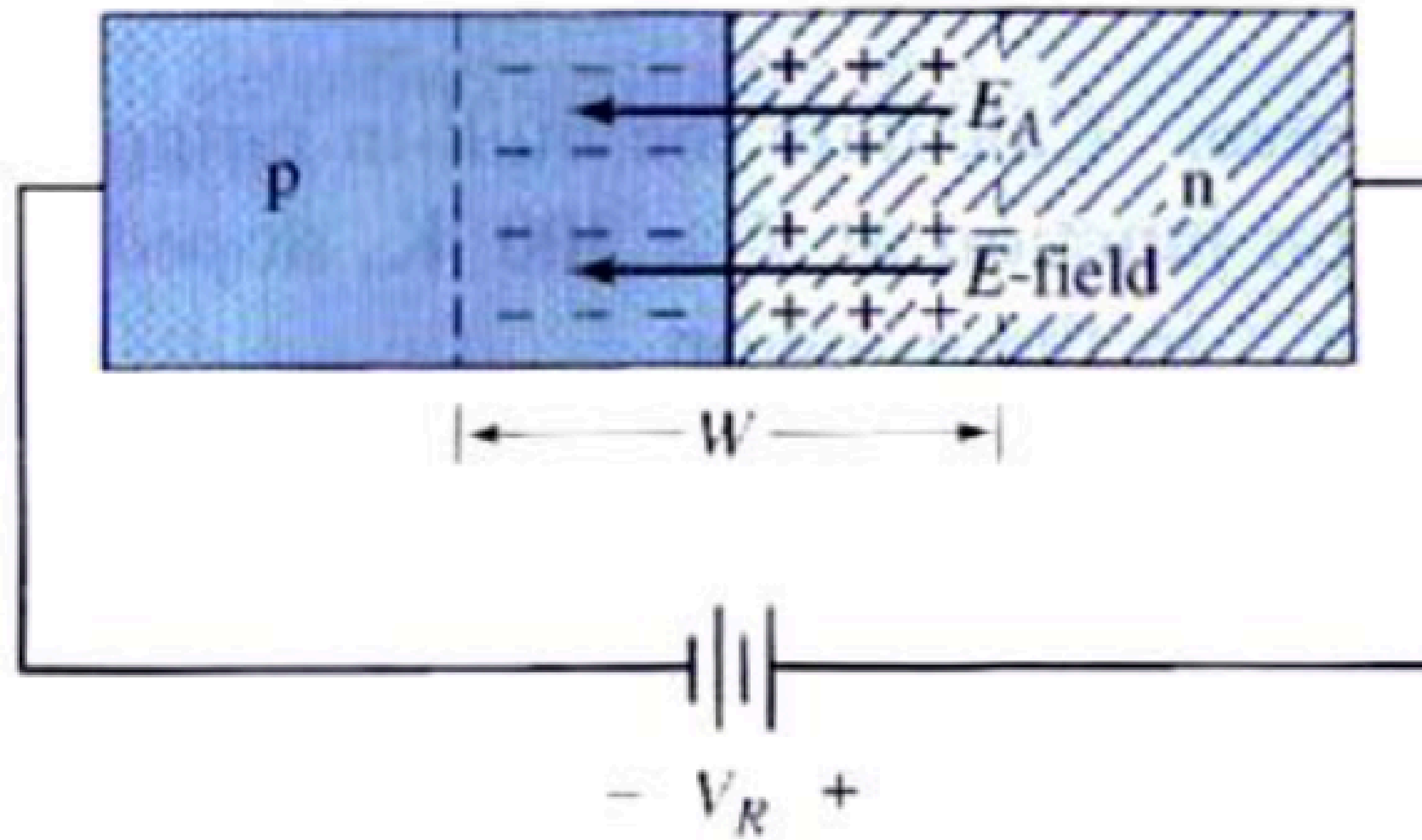


Working principles of diodes

Reverse-Biased PN Junction



Working principles of diodes

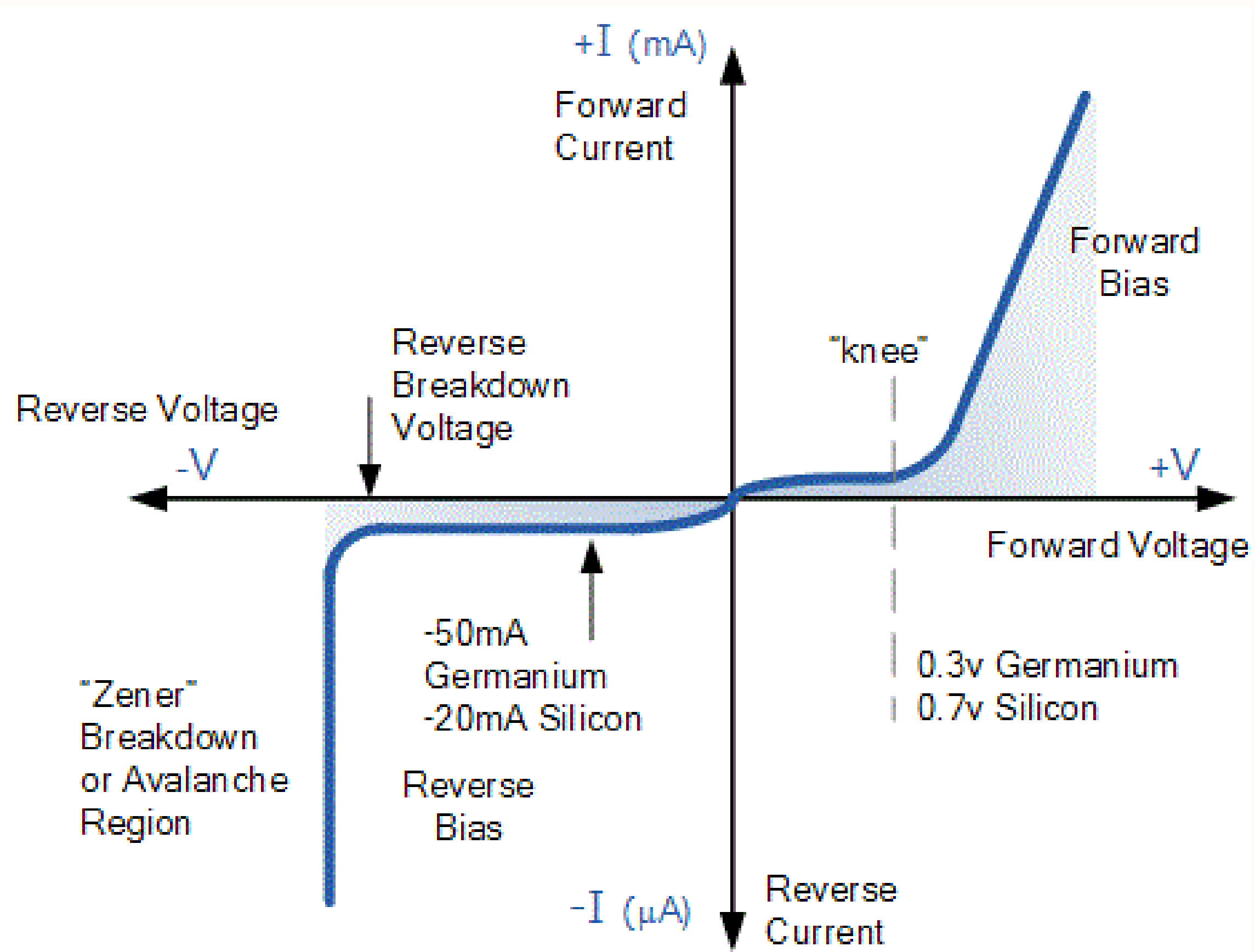


□ Reverse-biased PN junction

- Electric fields are generated from carrier diffusion and induced from outside supplied voltage V_R (E_A)
- Two E-fields have the same direction, which result in larger space-charge region
- Higher E-field holds back holes in p-region and electrons in n-region that no current across the junction

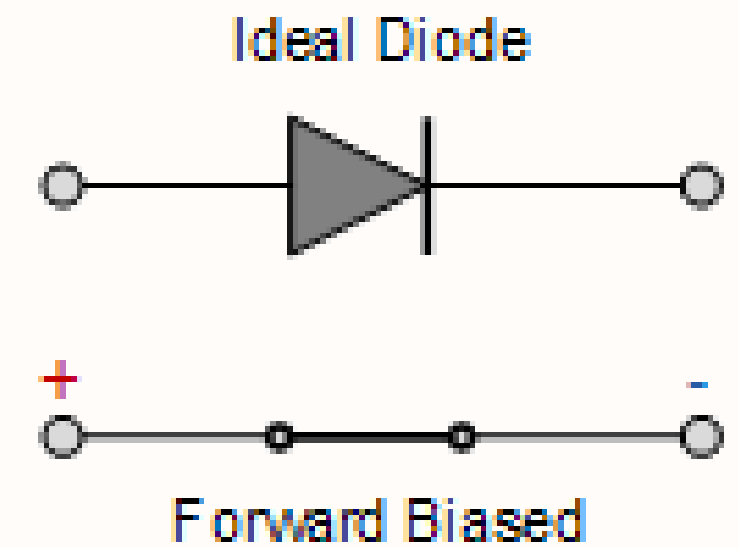
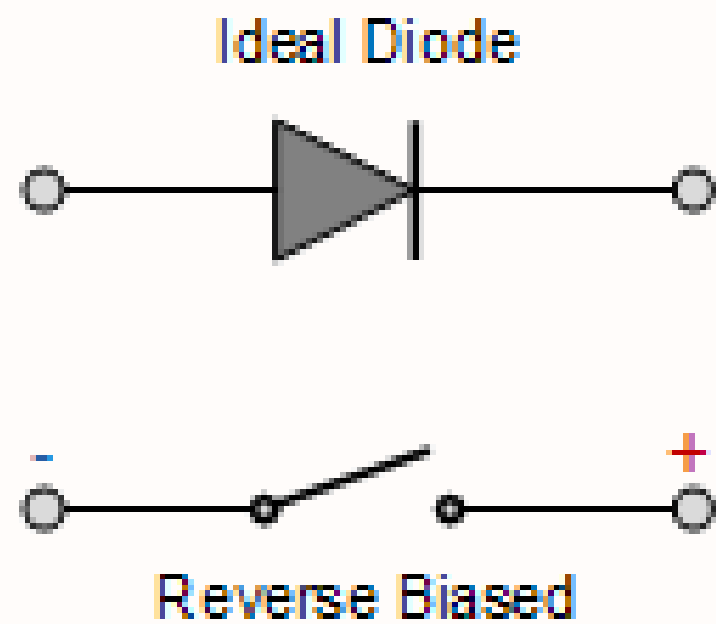
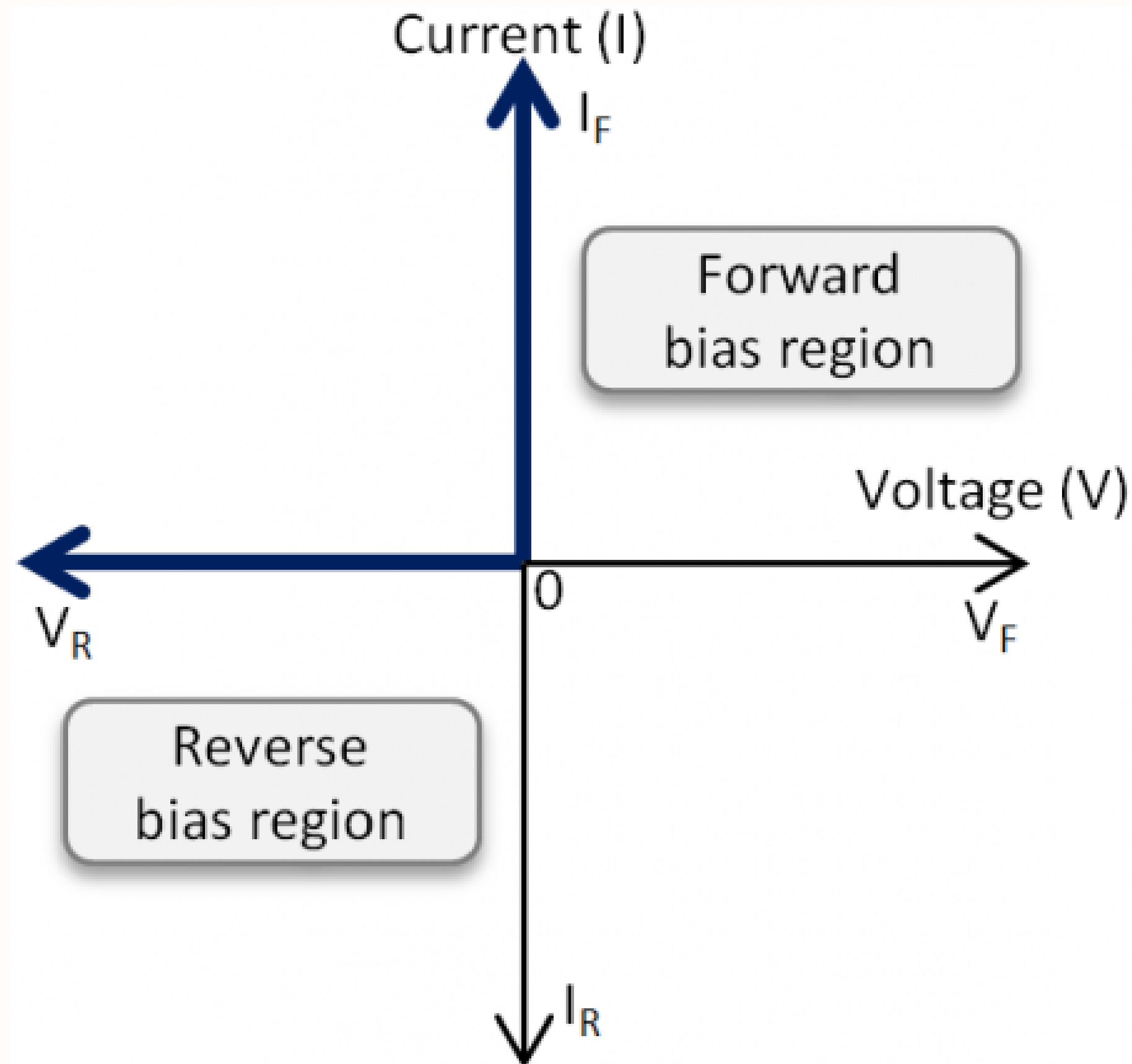


Diode I-V characteristic curve

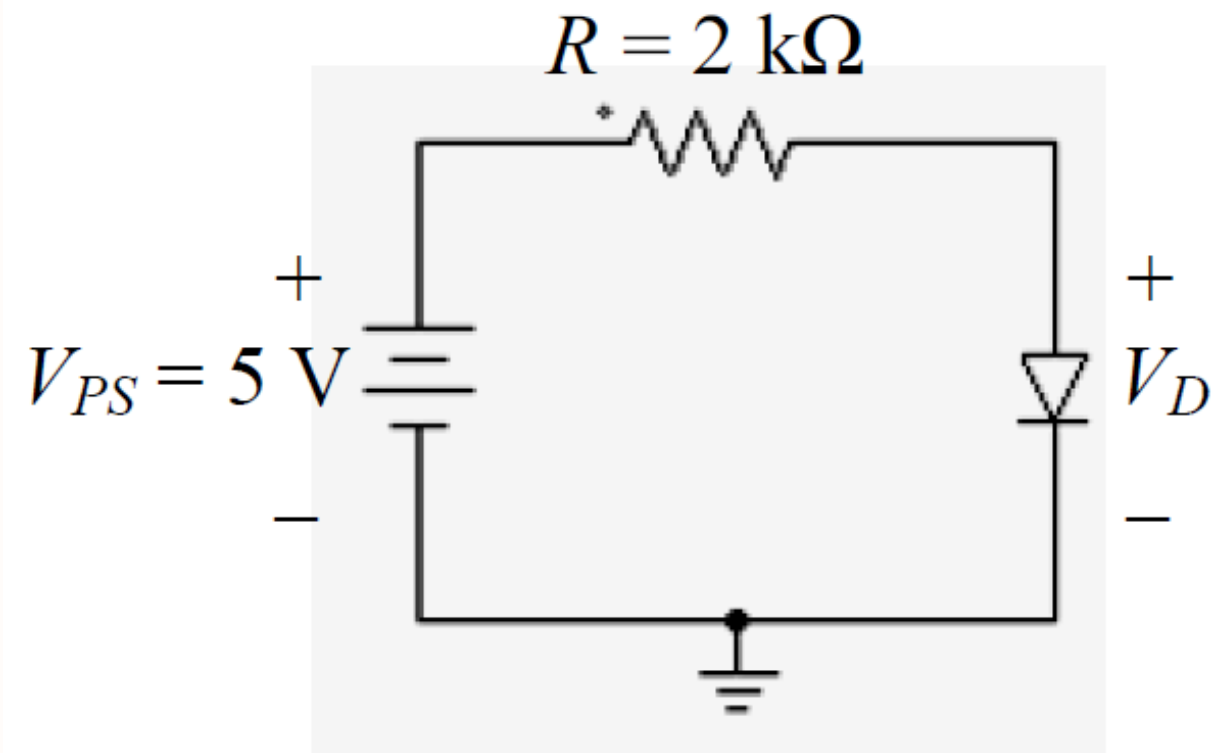




Ideal diode

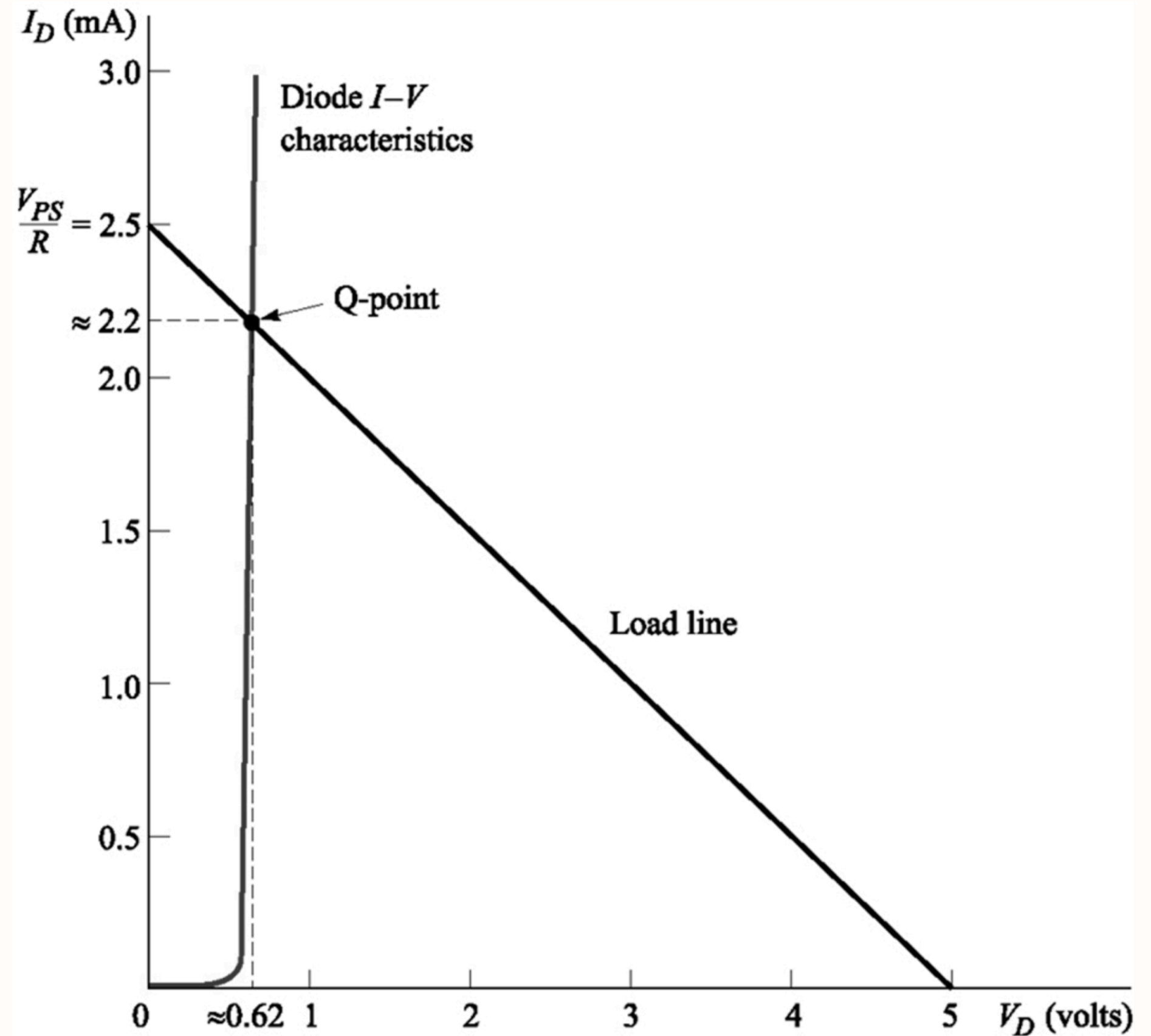


Practical diode circuit

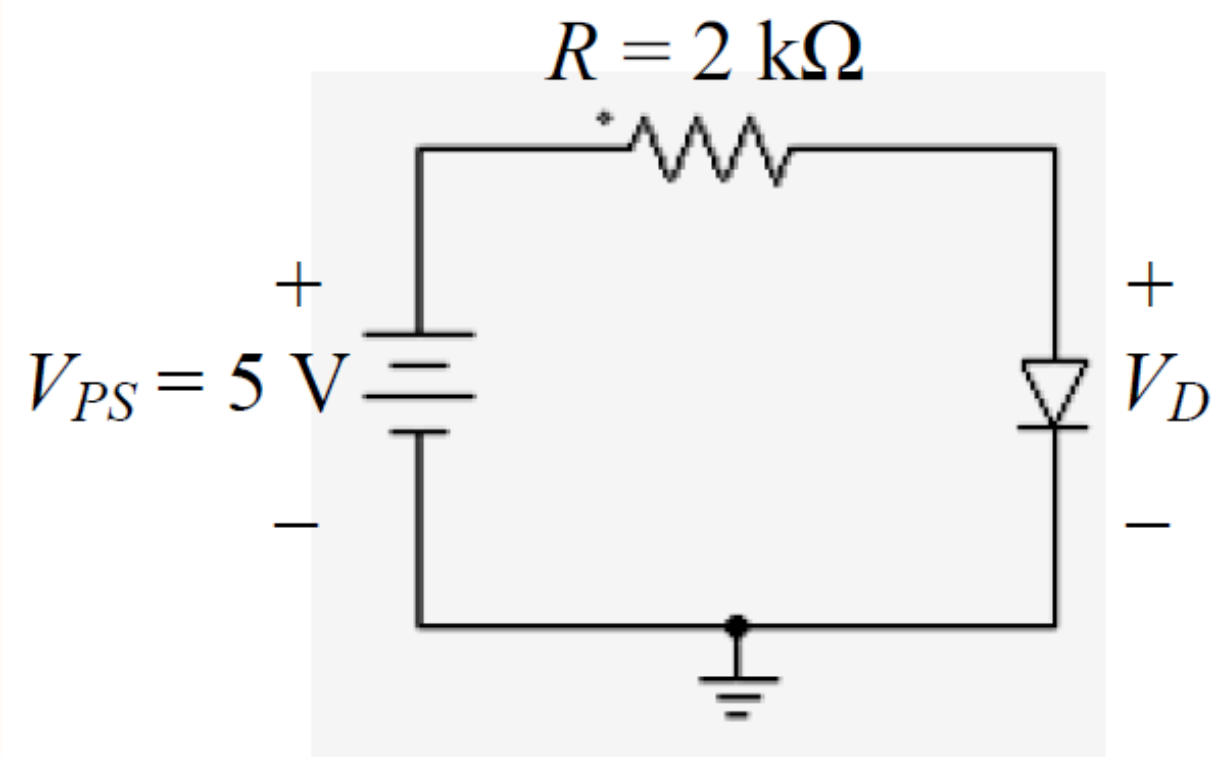


$$V_{PS} = I_D R + V_D$$

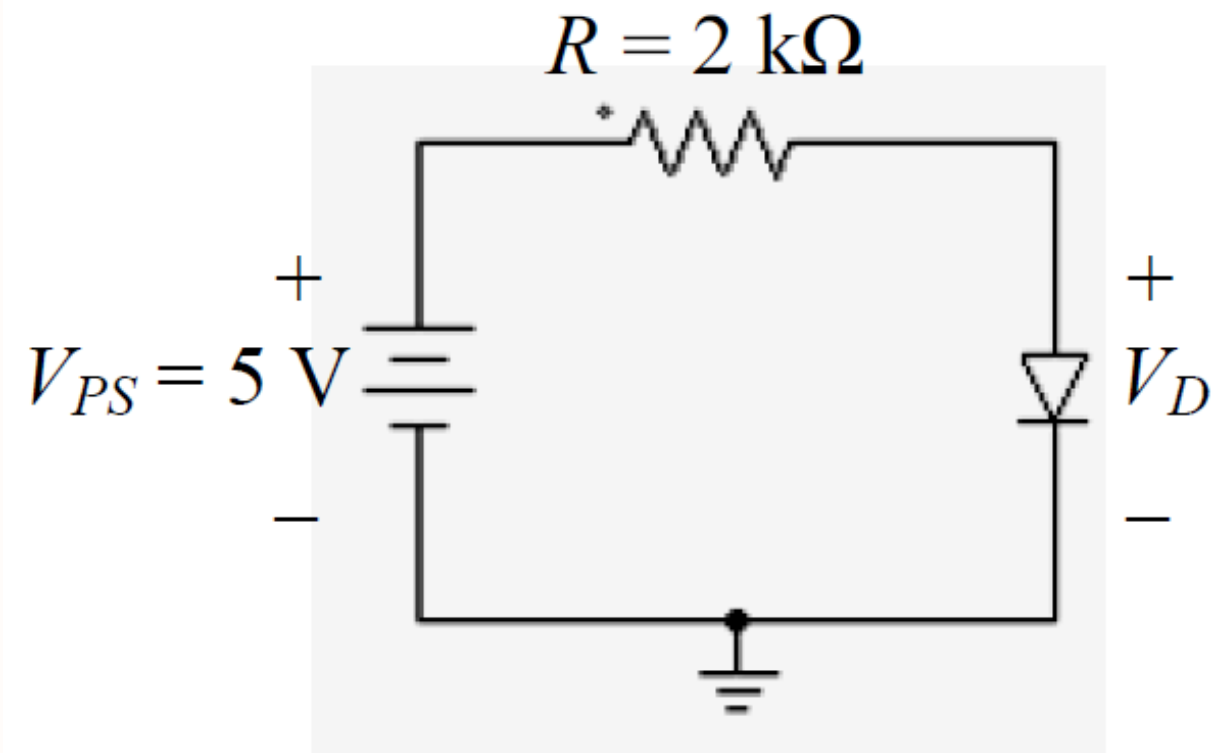
$$I_D = I_s \left[e^{\left(\frac{V_D}{nV_T} \right)} - 1 \right]$$



Diode circuit - Non-ideal case

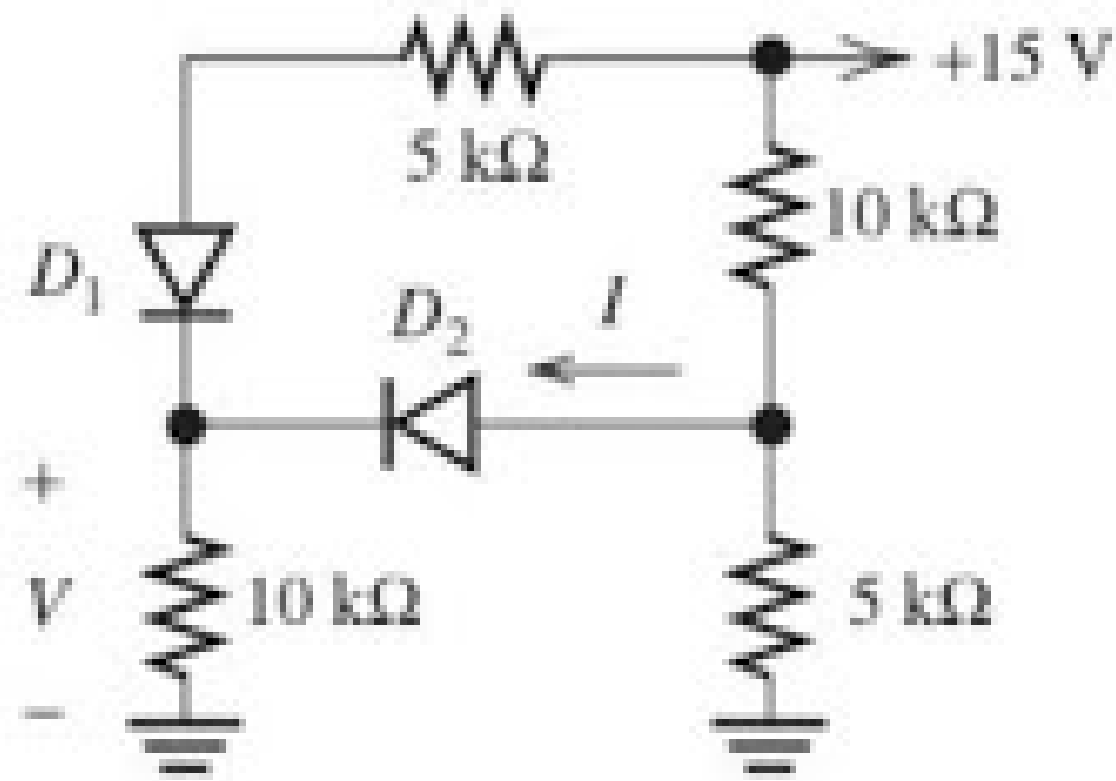


Diode circuit - Ideal case



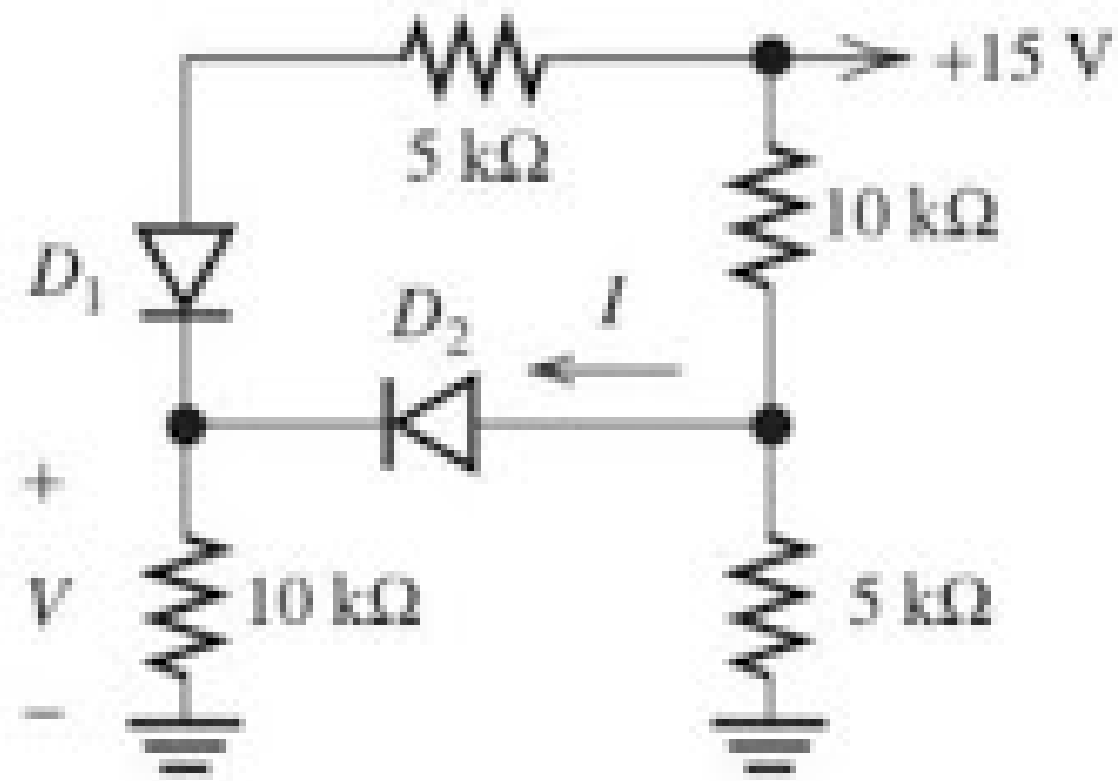


Non-ideal diode circuit - Example



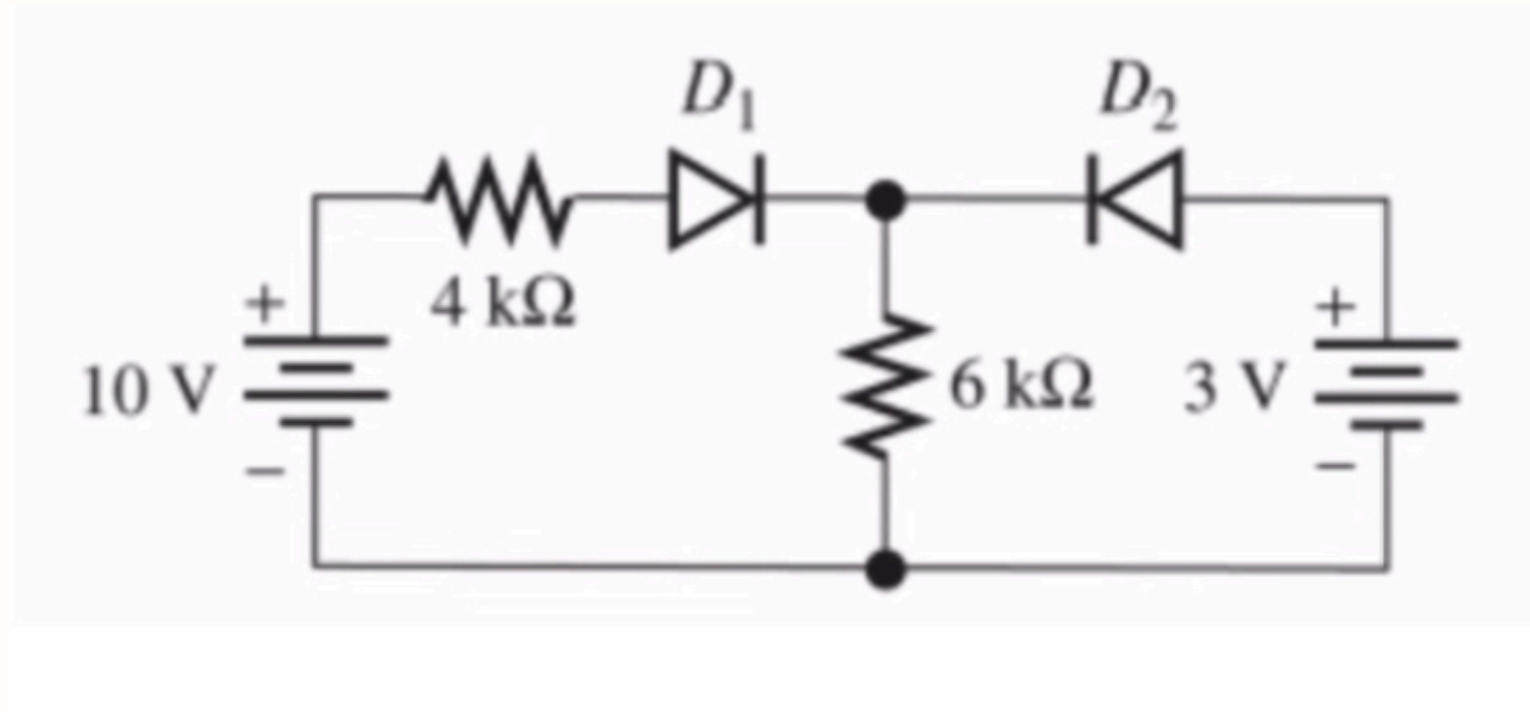


Ideal diode circuit - Example





Ideal-diode circuit - Assignment





Ideal-diode circuit - Assignment

